Resilience Loss of Tropical Forests in recent decades

Lana L. Blaschke[1,2*], Sebastian Bathiany[1,2], Marina Hirota[3], Niklas Boers[1,2,4]

[1]Munich Climate Center and Earth System Modelling Group, Department of Aerospace and Geodesy, TUM School of Engineering and Design, Technical University of Munich, Lise-Meitner-Straße 9, Ottobrunn, 85521, Bavaria, Germany.
[2]Complexity Science, Potsdam Institute for Climate Impact Research, Telegrafenberg A 31, Potsdam, 14473, Brandenburg, Germany.
[3]Department of Physics, Federal University of Santa Catarina, Florianópolis, Brazil.
[4]Department of Mathematics and Global Systems Institute, University of Exeter, Exeter, UK.

*Corresponding author(s). E-mail(s): lana.blaschke@tum.de;

Abstract

In view of ongoing climate and land use change, the resilience of tropical forests is crucial for maintaining ecosystem services and preventing potential forest loss and associated greenhouse gas emissions. Recent studies have attempted to quantify tropical forest resilience changes using different satellite vegetation products. However, it has not been assessed to what extent the data satisfies the theoretical assumptions of the employed resilience metrics. Here, we propose a framework to determine the most reliable combination of vegetation data and metrics to monitor resilience from space. We apply our framework to select the best combinations from 16 vegetation products and nine resilience metrics. Based on these, we consistently find that tropical forests in South America, Africa, and Asia have experienced a decline in resilience over recent decades. Our robust assessment of resilience changes in tropical forests has important implications for targeted actions to prevent further tropical forest loss.

Keywords: Tropical Forests, Resilience, Critical Slowing Down, Remote Sensing

1 Introduction

Tropical forests host a substantial fraction of species around the planet, and are a vital component of the global water and carbon cycles [1, 2]. However, anthropogenic climate and land use changes unprecedentedly threaten these ecosystems and put the services they provide at risk [3, 4]. In addition, it has been suggested that tropical forests might shift to an alternative low-tree-cover and degraded state due to a combination of directly human-driven and climatic disturbances [5–9]. In particular, the Amazon forest is widely recognised as a potential tipping element of the Earth system [10–12]. Moreover, the 2023 IPCC report rates the '[r]isks associated with large-scale singular events or tipping points, such as ice sheet instability or ecosystem loss from tropical forests, transition to high risk between 1.5°C to 2.5°C (medium confidence) and to very high risk between 2.5°C to 4°C (low confidence)' [13].

The mean state of a tipping element may remain constant as it approaches its tipping point. To still monitor its stability, numerous resilience investigations based on higher-order statistical properties have been conducted in recent decades for several proposed tipping elements [8, 12, 14]. In this context, remote sensing products play a key role in assessing and quantifying forest resilience [6, 8, 9, 12, 15]. Many satellites with an increasing variety of sensors have been launched for earth observation. However, it remains uncertain which remote sensing vegetation products (VP) are the most reliable [16, 17], and which should be used to investigate resilience in high-biomass tropical forests [15]. An approach to assess the pertinence of a VP to quantify and analyse resilience should ensure that the data used meet at least approximately the assumptions of the underlying theory.

Several studies [6, 8, 9, 12, 18] have used remote sensing data to assess the resilience of tropical forests, defined as the vegetation's ability to recover from disturbances. As a direct resilience estimate, the recovery rate can be estimated by fitting an exponential function to the recovery period after a large singular perturbation. However, this requires a significant perturbation within the observed time frame. Moreover, to measure changes of resilience over time, many such perturbations would have to occur. Alternatively, resilience can be estimated using dynamical systems theory. Here, we assume that vegetation dynamics are driven by dynamical noise that is induced by weather fluctuations and represented by a diffusion term. For simplicity, we will simply call this noise, implicitly assuming that the distinction to measurement noise is clear. Then, the system dynamics around equilibrium are modelled by an Ornstein-Uhlenbeck (OU) process, defined as

$$dx_t = \lambda x_t dt + \sigma dW_t \tag{1}$$

with a constant diffusion coefficient σ. The *recovery rate* λ determines the recovery speed to the mean, as the system is stable (mean-reverting) when $\lambda < 0$. If $\lambda > 0$, the system is unstable. Approaching $\lambda \nearrow 0$, recovery times lengthen, which is often referred to as *critical slowing down* (CSD). Changes in statistical proxies for λ form the basis for CSD resilience studies [19].

The two most commonly employed statistical proxies are the autocorrelation at lag 1 (AC1) and the variance (Methods). Although various other metrics have been

proposed and applied [20–23], AC1 is usually considered relatively robust and used by all the studies mentioned above. However, they differ with respect to their goal as well as the VP they use. The probably most well-known VP is the Normalized Vegetation Difference Index (NDVI) from the Advanced Very-High-Resolution Radiometer (AVHRR), continued by the Visible Infrared Imaging Radiometer Suite (VIIRS), or the Moderate Resolution Imaging Spectroradiometer (MODIS). The NDVI is an optical vegetation index using green and near-infrared frequency bands. Verbesselt et al. [6] employed the MODIS NDVI to infer resilience using AC1. Forzieri et al. [8] evaluated forest resilience change globally using the kNDVI, a supposedly more robust kernel version of NDVI [24]. However, Smith and Boers [15] investigated metric coherence in the MODIS Enhanced Vegetation Index (EVI), NDVI, kNDVI, a gross primary production index (GPP), and a Leaf Area Index. For evergreen forests, their results indicated issues in resilience analysis for all MODIS VPs, but especially EVI, NDVI, and kNDVI. In addition, Smith et al. [9] compared resilience metrics with empirically estimated recovery rates from perturbation recovery in NDVI from AVHRR and in a merged Vegetation Optical Depth (VOD) product, i.e. the Ku-band VOD from the Vegetation Optical Depth Climate Archive (VODCA) [25]. Smith et al. [9] found that neither AC1 nor variance effectively matches the empirical recovery rates for NDVI and suggested that recovery estimates are more accurate for VOD than for optical data. Therefore, Smith et al. [9] based their resilience analysis on the VODCA dataset, as did Boulton et al. [12]. Both studies revealed a significant decrease in resilience in tropical forests. Particularly in the Amazon, Boulton et al. [12] found large-scale increases of AC1. However, Smith et al. [26] highlighted potential biases due to non-stationarities of merged datasets influencing variance and autocorrelation. Consequently, single-sensor data should be preferred over merged data for reliable resilience estimation. Verbesselt et al. [6] had supported their findings using single-sensor data, and Blaschke et al. [18] based their analysis on four single-sensor VOD products (X- and C-VOD from the Advanced Microwave Scanning Radiometer - Earth Observing System sensor (AMSR-E) and C1- and C2-VOD from AMSR-E's successor AMSR2). Along with variance and AC1, the latter study assessed resilience by spatial correlation.

Tropical forest resilience studies are as numerous as ambiguous, mainly due to manifold uncertainties in the data, in the specific vegetation characteristics supposedly represented by a satellite product, and the suitability to quantify resilience via CSD. It is vital that data accurately represent the Earth's relevant features, such as aboveground biomass or vegetation water content. Furthermore, the applied resilience metrics imply that these features can be locally represented by an OU process. This raises significant challenges: theoretical assumptions must be followed by the VPs, but often even identifying the right variables is challenging. Thus, we see three ways to enhance the reliability of resilience analyses: first, a deeper understanding of ecological processes is essential to identify relevant variables. Second, this knowledge should guide us in choosing datasets that best represent the expected variable and vegetation function, and in choosing a resilience metric. Third, to ensure accuracy, we should verify whether the dataset meets the assumptions imposed by the resilience metric. Here we assess the third point, for which it is essential to remember that variance and AC1 as CSD-metrics rely on an underlying OU process. This implies that the

system is close to equilibrium. For tropical forests, this is a plausible approximation [27]. However, the OU hypothesis also imposes additional assumptions. First, both the recovery rate and the diffusion should be independent of the state, i.e. constants. Secondly, and with potentially larger effects on CSD-based resilience metrics, the noise dW_t of an OU should be Gaussian and white and its amplitude should be constant over time. In general, there is no direct way to ensure a time series originates from a particular dynamical system. Therefore, one cannot directly verify if a given dataset adheres to an OU process. Thus, we first verify each of the aforementioned assumptions as proposed by Bathiany et al. [17], before investigating by which metrics and using which dataset the resilience is most reliably measurable. To accomplish a comprehensive assessment, we compare 16 different remotely sensed vegetation products and nine resilience metrics.

In the following, we first assess for each VP separately if the theoretical assumptions are satisfied. Second, we perform a validation check by comparing whether the resilience metrics obtained match the empirical recovery rates following Smith et al. [9]. Both steps will guide us to an optimal set of VPs and metrics suitable to analyse the resilience of tropical vegetation. Based on the optimal combinations, we perform a reliable analysis of the change in resilience of tropical forests.

2 Results

We evaluate a wide range of satellite datasets that measure tropical vegetation (Methods). An overview of the monthly VPs analysed, the region, and the selection of grid cells excluding human influence are shown in Fig. 1. Besides the AC1 and variance, we test seven other resilience metrics that are based on CSD (see Methods).

2.1 Gaussian white noise with stationary amplitude

The state-space independence of the drift (see Sect. S1.1) and of the diffusion (see Sect. S1.2) for all datasets allows us to estimate the noise of the time series as the residuals of a generalised least squares regression with an AR covariance structure (GLSAR). To test whether the noise is Gaussian, we employ the Shapiro-Wilk test (Fig. 2a). Its low p-values for several datasets imply that the assumption of Gaussian noise is not valid in general. However, the higher p-values for the single-sensor VOD datasets indicate a higher probability of Gaussian noise for these datasets. This could hint at fewer problems caused by saturation and clouds compared to optical datasets. For details, we refer the reader to Sects. S1.4 and S3.1.

Moreover, if not taken care of, autocorrelated noise can affect AC1, which might give biased results when used as a metric of CSD. To test whether the noise is white, i.e. uncorrelated in time, we employ the Breusch-Godfrey test (Fig. 2b). The distribution of its p-values is particularly interesting: as only the X-VOD$_{\text{AMSR-E}}$ has a median p-value above 0.05, and basically all datasets are likely to have autocorrelated noise. For details, see Sects. S1.5 and S3.2.

Similarly, the amplitude of the noise affects the variance and, if changing, its reliability as a metric of resilience is affected. Thus, we retrieve noise amplitudes $\sigma(t)$ on sliding windows (Methods). Fig. 2c depicts the distribution of their Kendall τ.

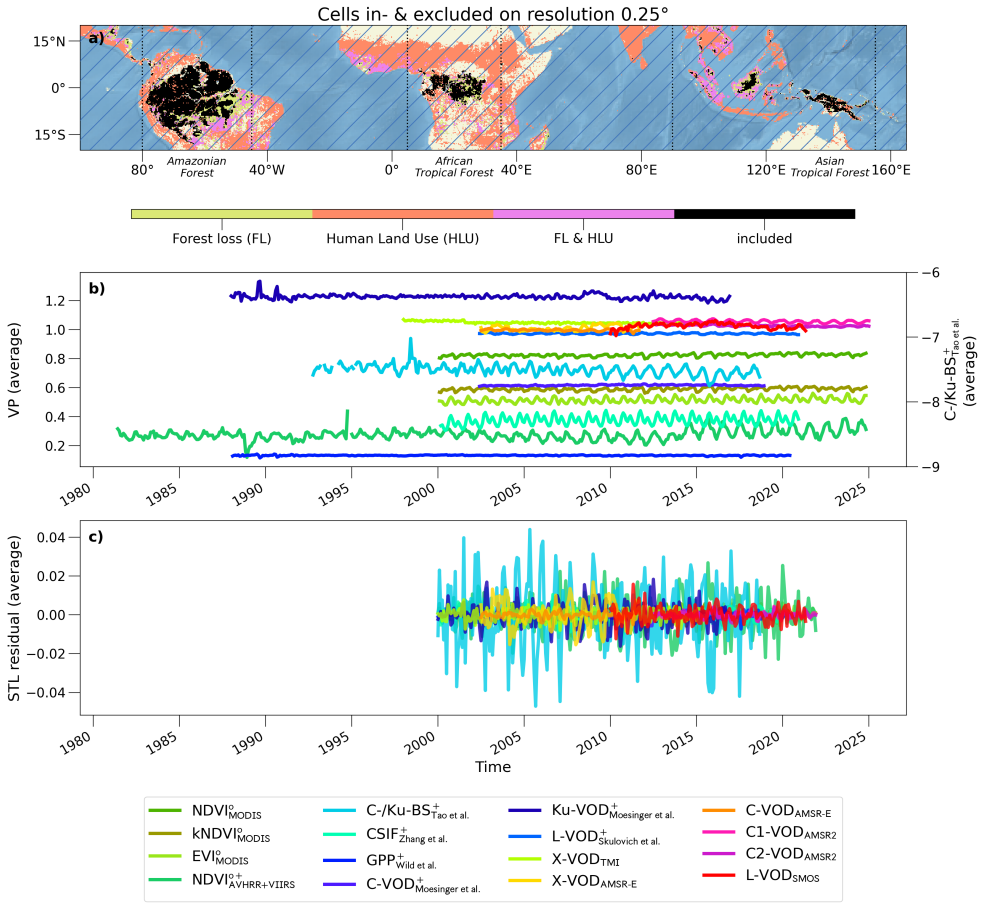

Fig. 1 Tropical forest areas and remote sensing vegetation indices. Panel a) shows the domain and the included and excluded areas, with data regridded to a 0.25° resolution. Hatched areas have less than 80 % evergreen vegetation. Panels b) and c) show the spatial average over the entire study region of the original and detrended Vegetation products, respectively. The analysed time period is restricted to the years 2000 to 2021 due to the availability of Human Land Use data. The superscripts o and + indicate that the data is optical and merged, respectively.

Remarkably, the average Kendall τ is further away from zero for any of the merged datasets compared to all other datasets, except L-VOD$^{+}_{\text{Skulovich et al.}}$ and X-VOD$_{\text{TMI}}$. This suggests that the merged products' biases originating in the differing signal-to-noise ratios, as shown by Smith et al. [26] for the VODCA data, can potentially be overcome by machine learning. Moreover, we find that all VPs include grid cells with strong changes in the noise amplitude, see Sects. S1.6 and S3.3 for details. To avoid any biases from changing noise amplitudes, we therefore generally advise to be careful about the use of uncorrected variance as a metric of CSD.

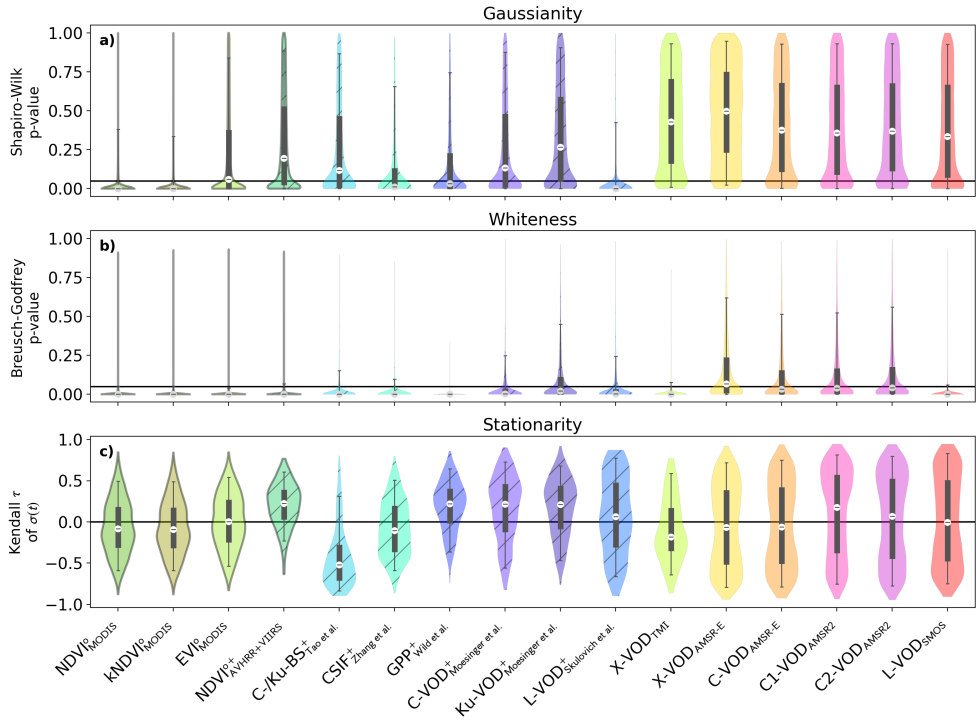

Fig. 2 Distribution of the Shapiro-Wilk and the Breusch-Godfrey tests' p-values as well as the change of the noise amplitude. A low p-value for a) the Shapiro-Wilk test indicates that the noise is unlikely to be Gaussian. Equivalently, in b) a low p-value for the Breusch-Godfrey test hints at a non-white noise process. In c), the noise is estimated on sliding windows of 5 years to retrieve $\sigma(t)$. The Kendall τ is employed as an indicator of the steadiness of change of $\sigma(t)$, with -1 indicating steady decrease and +1 a steady increase. Bars show the interquartile range and white dots the medians. The gray outline and the hatching of the violins indicate optical and merged data, respectively.

2.2 Empirical recovery in vegetation processes

Since none of the data meets all the theoretical assumptions of the OU hypothesis, it is not guaranteed that the statistical proxies of λ capture the features of recovery after larger perturbations in the vegetation variable [9]. We hence compare our resilience metrics to an empirical estimate of the recovery rate $\lambda_{\text{empirical}}$ for each dataset, following Smith et al. [9]. An example, and maps of $\lambda_{\text{empirical}}$ and its goodness-of-fit R^2 can be found in Sect. S2.

Fig. 3a shows the correlations of the different metrics with $\lambda_{\text{empirical}}$ for different goodness-of-fit levels, averaged over all VPs. For almost all metrics, the correlation coefficients increase prominently with increasing R^2. So, the more apparent an exponential recovery is, the better its rate is detected by the theoretical metrics. This means that the clearer the recovery signal, the better it can be captured by the theoretical metric, which validates the comparison as a tool to ensure that the theoretical metrics measure recovery. Moreover, Fig. 3a reveals that $\hat{\lambda}_{\text{GLSAR}}$ and AC1 perform

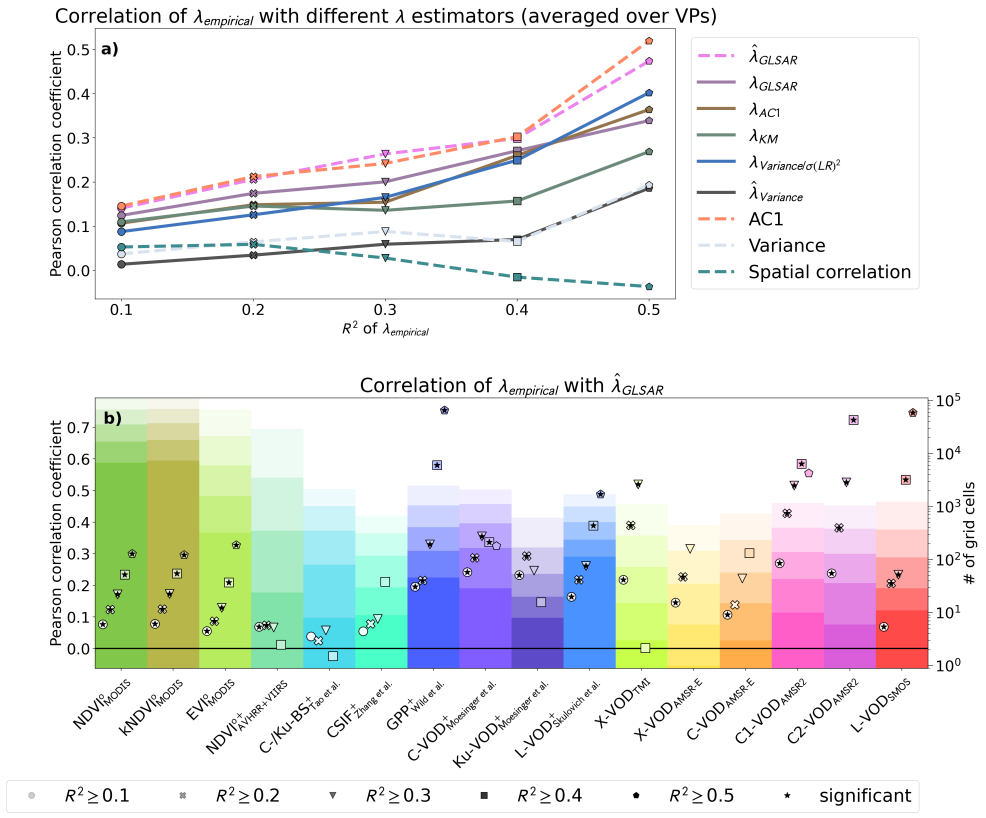

Fig. 3 The empirical recovery rate's correlation with its theoretical estimates a) averaged over all VPs, in dependence of the goodness-of-fit of the empirical recovery rate $\lambda_{\text{empirical}}$. Panel b) shows the Pearson correlation of $\hat{\lambda}_{\text{GLSAR}}$ separately for each VP. The different markers encode different goodness-of-fit levels, indicated by R^2. The amount of grid cells for which an empirical recovery rate of at least the given R^2, as well as the theoretical metric, could be calculated, is indicated by the histograms, shaded as the corresponding marker. Note their log-scaled y-axis on the right-hand side. A ★ indicates that the correlation is significant given the number of grid cells its calculated over.

best, independent of the correlation type (Sect. S2.2). In addition, it is striking that the variance and its $\hat{\lambda}_{\text{Variance}}$ are barely correlated with $\lambda_{\text{empirical}}$. This confirms that we cannot assume that the amplitude of the noise, σ, is similar across space. The spatial correlation is a function of spatially varying variables [18] and its poor correlation with $\lambda_{\text{empirical}}$ could result from such spatial differences. Unfortunately, we do not have an analytical formula to correct λ from the spatial correlation. Our approach is thus not suitable to test the reliability of spatial correlation and its trends can still be meaningful metrics of resilience change. Potentially due to its high data requirements and the limited length of the time series, λ_{KM} as suggested by Morr et al. [28] does not work well for the VPs studied here.

Fig. 3b shows the correlation of $\hat{\lambda}_{\text{GLSAR}}$ with $\lambda_{\text{empirical}}$ separately for each VP, allowing us to select data suitable to combine with $\hat{\lambda}_{\text{GLSAR}}$. All other metrics' results

can be found in Sects. S2.1 and S3.4. Regarding the optical datasets, three findings are noteworthy. First, our entire analysis could not reveal any difference in performance between $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ and $\text{kNDVI}^{\text{o}}_{\text{MODIS}}$, probably because the nature of the mapping of $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ into $\text{kNDVI}^{\text{o}}_{\text{MODIS}}$ does not introduce any changes that would matter to our statistical comparison. Second, $\text{EVI}^{\text{o}}_{\text{MODIS}}$ does not outperform $\text{NDVI}^{\text{o}}_{\text{MODIS}}$, which stands in contrast with claims about $\text{EVI}^{\text{o}}_{\text{MODIS}}$ being more suitable to improve the saturation curve for high biomass areas [29]. Third, $\text{NDVI}^{\text{o}+}_{\text{AVHRR+VIIRS}}$ shows very poor correlation. Furthermore, the merged data exhibit a wide spread in performance, with $\text{GPP}^{+}_{\text{Wild et al.}}$ and $\text{L-VOD}^{+}_{\text{Skulovich et al.}}$ on the upper end. For C-/Ku-$\text{BS}^{+}_{\text{Tao et al.}}$ and $\text{SIF}^{+}_{\text{Zhang et al.}}$, barely any perturbations were detected, for the latter probably due to its coarser spatial resolution compared to the other datasets. Regarding the single-sensor VOD datasets, $\text{X-VOD}_{\text{TMI}}$, $\text{X-VOD}_{\text{AMSR-E}}$ and $\text{C-VOD}_{\text{AMSR-E}}$ perform most poorly, the former supposedly due to its low signal-to-noise ratio [25]. For $\text{X-VOD}_{\text{AMSR-E}}$, we suspect that it likely inherits measurement errors (see Sect. S2.3). In general, the VPs that reveal the highest correlations between the empirical $\lambda_{\text{empirical}}$ and the theoretical $\hat{\lambda}_{\text{GLSAR}}$ are three single-sensor datasets, namely $\text{C1-VOD}_{\text{AMSR2}}$, $\text{C2-VOD}_{\text{AMSR2}}$, and $\text{L-VOD}_{\text{SMOS}}$ and two merged datasets, namely $\text{GPP}^{+}_{\text{Wild et al.}}$ and $\text{L-VOD}^{+}_{\text{Skulovich et al.}}$. This is especially true for the higher R^2 values, such that these results are even more trustworthy.

2.3 Resilience analysis

In the following, we evaluate resilience changes in tropical forests based on the five best-performing datasets, and additionally $\text{NDVI}^{\text{o}}_{\text{MODIS}}$, as the $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ correlates reasonably given its great amount of detectable $\lambda_{\text{empirical}}$ while providing single-sensor time series of the entire study period. Fig. 4 shows the linear trends of $\hat{\lambda}_{\text{GLSAR}}$ for the most reliable single-sensor VOD data, i.e. $\text{C1-VOD}_{\text{AMSR2}}$, $\text{C2-VOD}_{\text{AMSR2}}$, and $\text{L-VOD}_{\text{SMOS}}$. The latter dataset's time period does not completely match that of the AMSR2 data, thus, the findings are only comparable to a certain degree. $\text{C1-VOD}_{\text{AMSR2}}$ shows resilience loss in about half of the grid cells, with clusters around Iquitos, south of Manaus, in the northeast and northwest of the Amazonian Forest, a cluster around Kisangani in the African Forest, and the center of New Guinea for the Asian Forest. $\text{C2-VOD}_{\text{AMSR2}}$ shows resilience loss for more than half of the grid cells only in the Asian Forest. However, the clusters around Iquitos, south of Manaus, and in the Northwest of the Amazonian Forest as well in the center of New Guinea agree with that of $\text{C1-VOD}_{\text{AMSR2}}$. $\text{L-VOD}_{\text{SMOS}}$ shows decreasing resilience mainly in the central and western Amazonian Forest, along the northeastern border of the African Forest, and a cluster on Borneo and New Guinea. Fig. 5 shows the linear trends of $\hat{\lambda}_{\text{GLSAR}}$ for the data that is available across (almost) the entire study period, i.e. $\text{NDVI}^{\text{o}}_{\text{MODIS}}$, $\text{GPP}^{+}_{\text{Wild et al.}}$ and $\text{L-VOD}^{+}_{\text{Skulovich et al.}}$. $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ shows a majority of grid cells lost resilience for all three regions. Clusters of loss can be seen south of Iquitos and extensively around Manaus in the Amazon, as well as around Kisangani and along the northern border of the African Forest. $\text{GPP}^{+}_{\text{Wild et al.}}$ shows an even more pronounced resilience loss in all three regions, with patterns similar to $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ but being more pronounced also in the north of the Amazonian Forest, and pronounced signs of resilience loss in the Borneo forest. With 84.2 %, 77.2 %, and 76.3 % of cells with

decreasing resilience, the L-VOD$^{+}_{\text{Skulovich et al.}}$ reports widespread resilience loss across all three tropical forests. Note that all the changes are independent of measuring them as a trend or via Kendall τ, compare Sect. S2.4. The results for all excluded VPs can be found in Sect. S2.5.

Fig. 6 shows the spatially averaged time series of $\hat{\lambda}_{\text{GLSAR}}$ per region. For easier comparison, their range is adjusted to that of NDVI$^{\text{o}}_{\text{MODIS}}$. The original values as well as the results for the excluded VPs can be found in Sect. S2.6. For all three regions, the five sensors agree in resilience loss until approximately 2019 and an increase thereafter until the end of the study period. For the Amazonian and the African Forest the continuation of NDVI$^{\text{o}}_{\text{MODIS}}$, C1-VOD$_{\text{AMSR2}}$, and C2-VOD$_{\text{AMSR2}}$ until 2024 suggests we may be seeing the start of a new decline in resilience. Even though the continuation should be approached with caution, as we cannot guarantee that cells are free of human interference after 2021 (see Methods), the resilience loss seems plausible due to the occurrence of severe droughts (e.g., 2023 - 2024 in the Amazon [30]). Notably, whenever periods overlap, the different sensors yield remarkably consistent results. In particular, this is the case for NDVI$^{\text{o}}_{\text{MODIS}}$ with all other sensors in all three regions, except for L-VOD$_{\text{SMOS}}$ in Asia. However, the lower number of HLU-free grid cells and the numerous large water bodies in the area could explain the noiser signal and the disagreement of L-VOD$_{\text{SMOS}}$ for the Asian Forest. Originating in different missions, sensors, and frequency bands, their high agreement indicates that their $\hat{\lambda}_{\text{GLSAR}}$ indeed captures the dynamics of the underlying vegetation processes. Moreover, the agreement is weaker for the other VPs, see Sect. S2.6. This reconfirms our choice of combinations of data and metrics above, and provides evidence that our results as given in Figs. 5 and 4 are trustworthy. In particular, while biases from the merging cannot be ruled out entirely for GPP$^{+}_{\text{Wild et al.}}$ and L-VOD$^{+}_{\text{Skulovich et al.}}$ [26], their patterns and average evolution are well backed-up by NDVI$^{\text{o}}_{\text{MODIS}}$, and they might as well capture real and alarming resilience loss.

3 Discussion

Tropical forests are indispensable for climate change mitigation, but are at risk of shifting to a low-treecover state [5, 31, 32]. Several studies have investigated resilience changes in tropical forests, applying CSD-based resilience metrics to remotely sensed data. However, these studies hardly tested whether the theory employed and the properties of the (remotely sensed) data align, with only a few exceptions [9].

Here, we evaluated a wide range of VPs with respect to the assumptions underlying CSD-based resilience analyses. We first showed that most satellite vegetation datasets exhibit autocorrelated noise, potentially leading to biases in resilience analyses based on the common AC1 indicator. Moreover, a changing noise amplitude causes greater variation in the data and thus directly affects variance as a resilience metric. Here, particularly strong changes in the noise amplitude were found in the merged data. The correlation with empirical recovery rates confirmed $\hat{\lambda}_{\text{GLSAR}}$ as a robust resilience metric. Furthermore, the correlation analysis identified GPP$^{+}_{\text{Wild et al.}}$, L-VOD$^{+}_{\text{Skulovich et al.}}$, L-VOD$_{\text{SMOS}}$, C1-VOD$_{\text{AMSR2}}$, and C2-VOD$_{\text{AMSR2}}$ as the best data sets for consistent resilience estimation. Based on these and NDVI$^{\text{o}}_{\text{MODIS}}$, we assessed

resilience changes in terms of $\hat{\lambda}_{\mathrm{GLSAR}}$. We found decreasing resilience until around 2019, with consensus among sensors for the Amazon and African forests. This agrees with the forests' declining ability to sequester carbon [2]. Except for L-VOD$_{\mathrm{SMOS}}$ and, partially, GPP$^{+}_{\text{Wild et al.}}$ in Asia, NDVI$^{\circ}_{\mathrm{MODIS}}$'s $\hat{\lambda}_{\mathrm{GLSAR}}$ time series, which spans the entire study period, exhibits a remarkable consistency with any of the other VPs' $\hat{\lambda}_{\mathrm{GLSAR}}$ during overlapping intervals, confirming the robustness of our study.

This study investigated datasets used in resilience studies of tropical forests in recent years. We will now discuss the reliability of previous results depending on the employed data-metric combinations. In general, we suggest that machine learning might have the potential of merging data while bypassing the concerns from Smith et al. [26] as L-VOD$^{+}_{\text{Skulovich et al.}}$ performed robustly in our comparisons. In particular, we could reproduce the findings by Smith et al. [9], showing that the theoretically derived recovery rates do not compare well to the empirical recovery rates for NDVI$^{\circ+}_{\mathrm{AVHRR+VIIRS}}$. There, the authors suggest that this could be a general problem of optical VPs, as a fast and green post-perturbation establishment of a new fast-growing species can show up as full recovery to optical sensors [17]. Yet, our findings for NDVI$^{\circ+}_{\mathrm{AVHRR+VIIRS}}$ do not transfer to the other optical VPs. Interestingly, even though NDVI$^{\circ}_{\mathrm{MODIS}}$, kNDVI$^{\circ}_{\mathrm{MODIS}}$, and EVI$^{\circ}_{\mathrm{MODIS}}$ do not meet the white noise assumption, their AC1 correlates well with the empirical recovery rate. Thus, our study does confirm the results of the CSD-based resilience analyses by Verbesselt et al. [6] and Forzieri et al. [8] using NDVI$^{\circ}_{\mathrm{MODIS}}$ and kNDVI$^{\circ}_{\mathrm{MODIS}}$, respectively. We note, however, that even though Forzieri et al. [8] claim that kNDVI is supposed to outperform NDVI in densely vegetated regions, both VPs' results are almost identical. This suggests that for CSD-based resilience analyses, the kernel-based NDVI (kNDVI) is unlikely to outperform NDVI. In comparison, for Ku-VOD$^{+}_{\text{Moesinger et al.}}$ neither λ_{AC1} nor $\lambda_{\mathrm{Variance}/\sigma(LR)^2}$ correlate significantly with empirical recovery in most R^2-categories. Our results appear inconsistent with those of Smith et al. [9], but time period, time resolution, and study area differ. Furthermore, their algorithm for $\lambda_{\mathrm{empirical}}$ was slightly adapted (Methods). Smith and Boers [15] found low coherence between λ_{AC1} and $\lambda_{\mathrm{Variance}/\sigma(LR)^2}$ for evergreen broadleaf forests, suggesting that vegetation resilience estimates' reliability for optical data depends on biomass density. We assert that certain VPs, e.g. NDVI$^{\circ}_{\mathrm{MODIS}}$, remain effective for particular resilience metrics, potentially depending on their gap-filling and deseasoning [33]. This emphasizes the critical importance of a careful selection of VPs and resilience metrics in these areas. Blaschke et al. [18] analysed X-VOD$_{\mathrm{AMSR\text{-}E}}$, C-VOD$_{\mathrm{AMSR\text{-}E}}$, C1-VOD$_{\mathrm{AMSR2}}$, and C2-VOD$_{\mathrm{AMSR2}}$ using variance, AC1, and spatial correlation as resilience metrics. According to Fig. S2.16, AC1 should be reliable metric particularly for C1-VOD$_{\mathrm{AMSR2}}$ and C2-VOD$_{\mathrm{AMSR2}}$, and variance for C1-VOD$_{\mathrm{AMSR2}}$. Even though all the data's noise might be autocorrelated, potentially influencing the AC1, only changes in autocorrelation strength would influence AC1 trends. Moreover, we find rather constant noise amplitudes in these single-sensor VOD data, hence, changes in variance should not be biased. As mentioned earlier, our approach does not allow us to validate the spatial correlation's reliability. However, the spatial patterns of resilience change shown in [18] compare well with ours, with agreement on resilience loss in the west, southeast,

and north. Overall, there is hence strong consensus suggesting these areas are losing resilience, necessitating focused efforts to prevent forest loss.

In summary, we introduced a comprehensive approach to close an urgent gap in the literature: verify the foundational assumptions of CSD-based resilience analyses in the data and confirm whether the theoretical metrics align with the real-world concept of recovery. This strategy allows a well-informed selection of an ideal combination of data and metric, and hence a reliable and robust assessment of resilience changes. We applied this approach to an extensive set of remotely sensed VPs to identify the best combination of VPs and resilience metrics. This combination was then used to perform a resilience analysis for the three tropical forest regions. In particular, its application to an extensive set of VPs revealed that for the vast majority, the assumptions of white and Gaussian noise with a constant amplitude are problematic and thus we considered $\hat{\lambda}_{\mathrm{GLSAR}}$, which is robust against changing variance and correlation of the noise, to be the most reliable resilience metric. We advise that future resilience research should apply the concept in order to reliably monitor possible resilience changes of the Earth's potential Tipping Elements. A better understanding of the system's dynamics and data that capture the relevant features of the system will eventually lead to enhanced reliability of resilience analyses. In particular, a closer collaboration between resilience scientists and remote sensing specialists has the potential to improve the interpretation of resilience study results and more accurate outcomes.

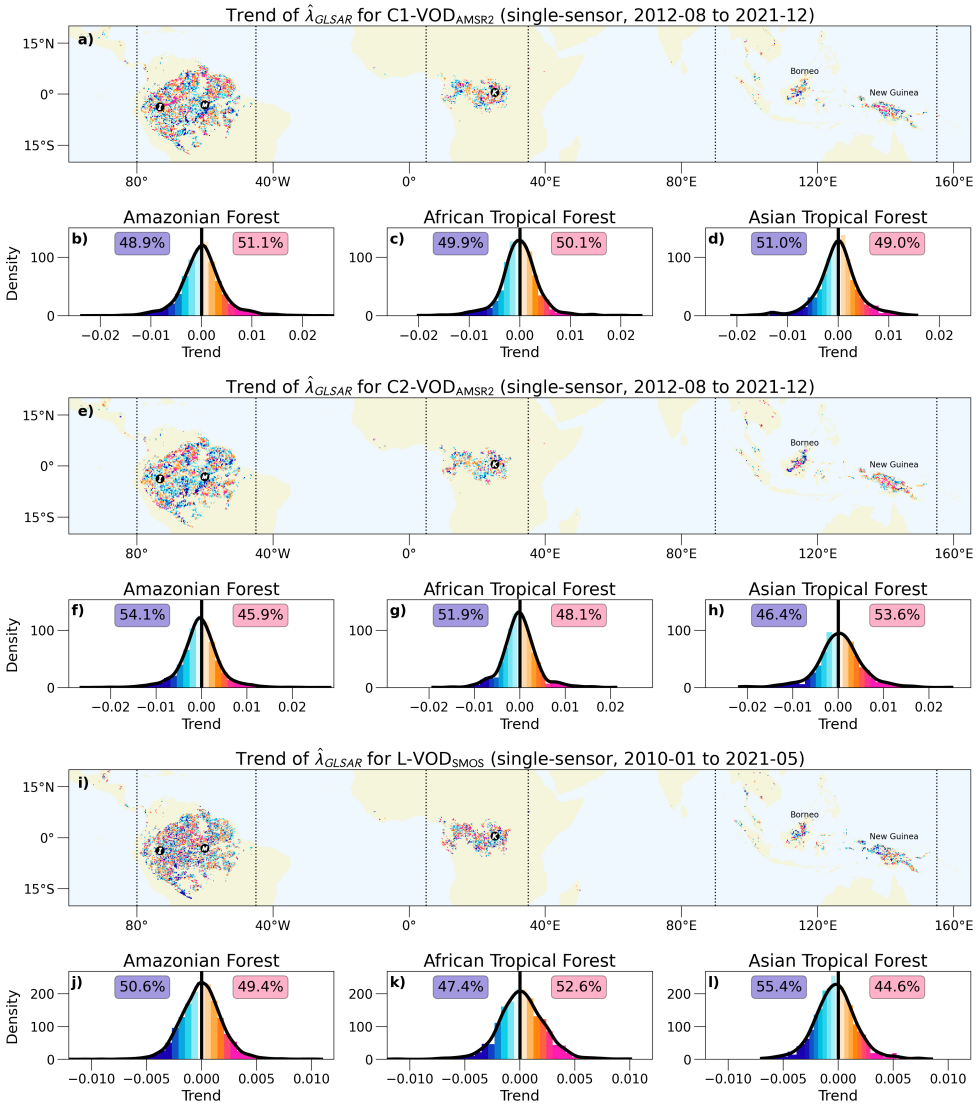

Fig. 4 **Resilience change assessed as trend** of $\hat{\lambda}_{\mathrm{GLSAR}}$, calculated on sliding windows of 5 years for the three single-sensor VOD data sets, i.e. a-d) C1-VOD$_{\mathrm{AMSR2}}$, e-h) C2-VOD$_{\mathrm{AMSR2}}$, and i-l) L-VOD$_{\mathrm{SMOS}}$. The maps a,e) and i) show the spatial distribution of the linear trend with colour-coding equivalent to the corresponding histograms, which are given separately for the three tropical regions. Namely, red (blue) codes increasing (decreasing) $\hat{\lambda}_{\mathrm{GLSAR}}$ and thus, resilience loss (gain). The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

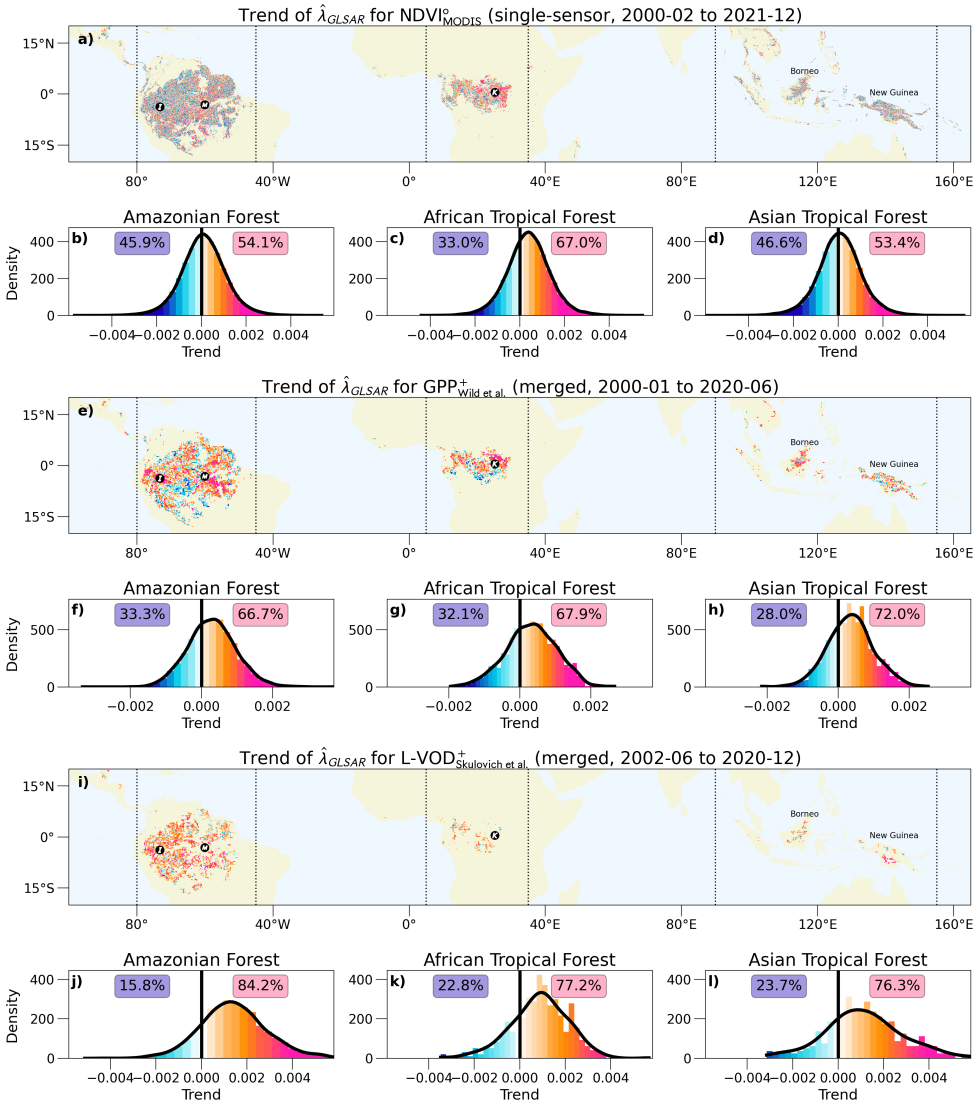

Fig. 5 **Resilience change assessed as trend** of $\hat{\lambda}_{\mathrm{GLSAR}}$, calculated on sliding windows of 5 years for the three datasets available on (almost) the entire study period, i.e. a-d) $\mathrm{NDVI}^{\mathrm{o}}_{\mathrm{MODIS}}$, e-h) $\mathrm{GPP}^{+}_{\text{Wild et al.}}$, and i-l) $\text{L-VOD}^{+}_{\text{Skulovich et al.}}$. The maps a,e) and i) show the spatial distribution of the linear trend with colour-coding equivalent to the corresponding histograms, which are given separately for the three tropical regions. Namely, red (blue) codes increasing (decreasing) $\hat{\lambda}_{\mathrm{GLSAR}}$ and thus, resilience loss (gain). The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

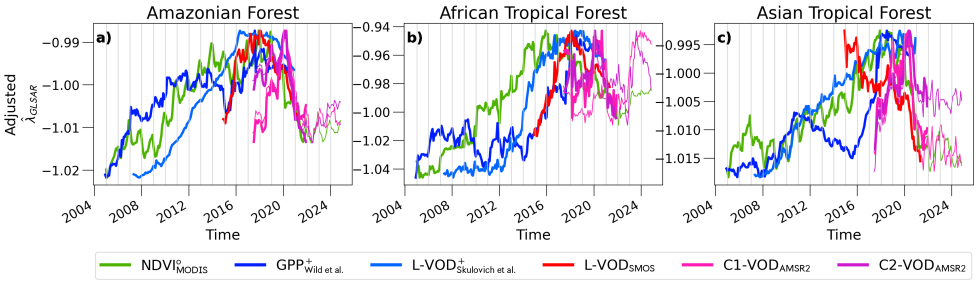

Fig. 6 Spatially averaged time series of $\hat{\lambda}_{\mathbf{GLSAR}}$ calculated on sliding windows of 5 years for the three tropical regions, i.e. a) the Amazon, b) the African and c) the Asian Tropical Forest. An increasing $\hat{\lambda}_{\mathrm{GLSAR}}$ signifies resilience loss. As indicated by the colors, they are based on $\mathrm{NDVI}^{\circ}_{\mathrm{MODIS}}$, $\mathrm{GPP}^{+}_{\text{Wild et al.}}$, $\mathrm{L\text{-}VOD}^{+}_{\text{Skulovich et al.}}$, $\mathrm{L\text{-}VOD}_{\mathrm{SMOS}}$, $\mathrm{C1\text{-}VOD}_{\mathrm{AMSR2}}$, and $\mathrm{C2\text{-}VOD}_{\mathrm{AMSR2}}$. The range in the latter ones is adjusted to fit that of $\mathrm{NDVI}^{\circ}_{\mathrm{MODIS}}$ over the same period for better comparison. Thick lines indicate results for the main time period from 2000-2021. For results until 2024 (thin lines), human land use in the last three years cannot be excluded.

References

[1] Salati, E., Dall'Olio, A., Matsui, E. & Gat, J. R. Recycling of water in the Amazon Basin: An isotopic study. *Water Resources Research* **15**, 1250–1258 (1979). URL https://onlinelibrary.wiley.com/doi/abs/10.1029/WR015i005p01250.

[2] Hubau, W. *et al.* Asynchronous carbon sink saturation in African and Amazonian tropical forests. *Nature* **579**, 80–87 (2020). URL https://www.nature.com/articles/s41586-020-2035-0. Number: 7797 Publisher: Nature Publishing Group.

[3] Allen, C. D. *et al.* A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *Forest Ecology and Management* **259**, 660–684 (2010). URL https://linkinghub.elsevier.com/retrieve/pii/S037811270900615X. Publisher: Elsevier BV.

[4] Lovejoy, T. E. & Nobre, C. Amazon Tipping Point. *Science Advances* **4**, eaat2340 (2018). URL https://advances.sciencemag.org/content/4/2/eaat2340. Publisher: American Association for the Advancement of Science Section: Editorial.

[5] Hirota, M., Holmgren, M., Van Nes, E. H. & Scheffer, M. Global Resilience of Tropical Forest and Savanna to Critical Transitions. *Science* **334**, 232–235 (2011). URL https://www.science.org/doi/10.1126/science.1210657.

[6] Verbesselt, J. *et al.* Remotely sensed resilience of tropical forests. *Nature Climate Change* **6**, 1028–1031 (2016). URL https://www.nature.com/articles/nclimate3108.

[7] Wunderling, N., Wolf, F., Tuinenburg, O. A. & Staal, A. Network motifs shape distinct functioning of Earth's moisture recycling hubs. *Nature Communications* **13**, 6574 (2022). URL https://www.nature.com/articles/s41467-022-34229-1. Number: 1 Publisher: Nature Publishing Group.

[8] Forzieri, G., Dakos, V., McDowell, N. G., Ramdane, A. & Cescatti, A. Emerging signals of declining forest resilience under climate change. *Nature* (2022). URL https://www.nature.com/articles/s41586-022-04959-9.

[9] Smith, T., Traxl, D. & Boers, N. Empirical evidence for recent global shifts in vegetation resilience. *Nature Climate Change* **12**, 477–484 (2022). URL https://www.nature.com/articles/s41558-022-01352-2. Number: 5 Publisher: Nature Publishing Group.

[10] Lenton, T. M. *et al.* Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences* **105**, 1786–1793 (2008). URL http://www.pnas.org/cgi/doi/10.1073/pnas.0705414105.

[11] Boers, N., Ghil, M. & Stocker, T. F. Theoretical and paleoclimatic evidence for abrupt transitions in the Earth system. *Environmental Research Letters*

17, 093006 (2022). URL https://iopscience.iop.org/article/10.1088/1748-9326/ac8944.

[12] Boulton, C. A., Lenton, T. M. & Boers, N. Pronounced loss of Amazon rainforest resilience since the early 2000s. *Nature Climate Change* **12**, 271–278 (2022). URL https://www.nature.com/articles/s41558-022-01287-8.

[13] Calvin, K. *et al.* IPCC, 2023: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland. Tech. Rep., Intergovernmental Panel on Climate Change (IPCC) (2023). URL https://www.ipcc.ch/report/ar6/syr/. Edition: First.

[14] Boers, N. & Rypdal, M. Critical slowing down suggests that the western Greenland Ice Sheet is close to a tipping point. *Proceedings of the National Academy of Sciences* **118**, e2024192118 (2021). URL https://www.pnas.org/doi/abs/10.1073/pnas.2024192118. Publisher: Proceedings of the National Academy of Sciences.

[15] Smith, T. & Boers, N. Reliability of vegetation resilience estimates depends on biomass density. *Nature Ecology & Evolution* 1–10 (2023). URL https://www.nature.com/articles/s41559-023-02194-7. Publisher: Nature Publishing Group.

[16] Konings, A. G., Rao, K. & Steele-Dunne, S. C. Macro to micro: microwave remote sensing of plant water content for physiology and ecology. *New Phytologist* **223**, 1166–1172 (2019). URL https://onlinelibrary.wiley.com/doi/abs/10.1111/nph.15808. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/nph.15808.

[17] Bathiany, S. *et al.* Ecosystem Resilience Monitoring and Early Warning Using Earth Observation Data: Challenges and Outlook. *Surveys in Geophysics* (2024). URL https://doi.org/10.1007/s10712-024-09833-z.

[18] Blaschke, L. L. *et al.* Spatial Correlation Increase in Single-Sensor Satellite Data Reveals Loss of Amazon Rainforest Resilience. *Earth's Future* **12**, e2023EF004040 (2024). URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023EF004040.

[19] Scheffer, M. *et al.* Early-warning signals for critical transitions. *Nature* **461**, 53–59 (2009). URL https://www.nature.com/articles/nature08227. Number: 7260 Publisher: Nature Publishing Group.

[20] Boers, N. Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nature Climate Change* **11**, 680–688 (2021). URL https://www.nature.com/articles/s41558-021-01097-4.

[21] Dakos, V., van Nes, E. H., Donangelo, R., Fort, H. & Scheffer, M. Spatial correlation as leading indicator of catastrophic shifts. *Theoretical Ecology* **3**, 163–174 (2010). URL https://doi.org/10.1007/s12080-009-0060-6.

[22] Kéfi, S. *et al.* Early Warning Signals of Ecological Transitions: Methods for Spatial Patterns. *PLoS ONE* **9**, e92097 (2014). URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3962379/.

[23] Tirabassi, G. Linear theory of the spatial signatures of critical slowing down. *Physical Review Research* **6**, 023228 (2024). URL https://link.aps.org/doi/10.1103/PhysRevResearch.6.023228.

[24] Camps-Valls, G. *et al.* A unified vegetation index for quantifying the terrestrial biosphere. *Science Advances* **7**, eabc7447 (2021). URL https://www.science.org/doi/10.1126/sciadv.abc7447. Publisher: American Association for the Advancement of Science.

[25] Moesinger, L. *et al.* The global long-term microwave Vegetation Optical Depth Climate Archive (VODCA). *Earth System Science Data* **12**, 177–196 (2020). URL https://essd.copernicus.org/articles/12/177/2020/. Publisher: Copernicus GmbH.

[26] Smith, T. *et al.* Reliability of resilience estimation based on multi-instrument time series. *Earth System Dynamics* **14**, 173–183 (2023). URL https://esd.copernicus.org/articles/14/173/2023/.

[27] Fan, N. *et al.* Implications of the steady-state assumption for the global vegetation carbon turnover. *Environmental Research Letters* **18**, 104036 (2023). URL https://iopscience.iop.org/article/10.1088/1748-9326/acfb22/meta. Publisher: IOP Publishing.

[28] Morr, A., Riechers, K., Gorjão, L. R. & Boers, N. Anticipating critical transitions in multidimensional systems driven by time- and state-dependent noise. *Physical Review Research* **6**, 033251 (2024). URL https://link.aps.org/doi/10.1103/PhysRevResearch.6.033251.

[29] Huete, A. *et al.* Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment* **83**, 195–213 (2002). URL https://www.sciencedirect.com/science/article/pii/S0034425702000962.

[30] Souza, C. M. *et al.* Amazon severe drought in 2023 triggered surface water loss. *Environmental Research: Climate* **3**, 041002 (2024). URL https://iopscience.iop.org/article/10.1088/2752-5295/ad7c71.

[31] Malhi, Y. *et al.* Climate Change, Deforestation, and the Fate of the Amazon. *Science* **319**, 169–172 (2008). URL https://www.sciencemag.org/lookup/doi/10.1126/science.1146961.

[32] Zemp, D. C. *et al.* Self-amplified Amazon forest loss due to vegetation-atmosphere feedbacks. *Nature Communications* **8**, 14681 (2017). URL http://www.nature.com/articles/ncomms14681.

[33] Liu, T. *et al.* The influence of data gaps and outliers on resilience indicators (2025). URL http://arxiv.org/abs/2505.19034. ArXiv:2505.19034 [nlin].

[34] Vermote, E. & NOAA CDR Program. NOAA Climate Data Record (CDR) of AVHRR Normalized Difference Vegetation Index (NDVI), Version 5 (2018). URL https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01558/html.

[35] Didan, K. MODIS/Terra Vegetation Indices Monthly L3 Global 0.05Deg CMG V061 (2021). URL https://lpdaac.usgs.gov/products/mod13c2v061/.

[36] Vrije Universiteit Amsterdam (Richard de Jeu) and NASA GSFC (Manfred Owe). AMSR-E/Aqua surface soil moisture (LPRM) L3 1 day 25 km x 25 km descending V002 (2011).

[37] Vrije Universiteit Amsterdam (Richard de Jeu) and NASA GSFC (Manfred Owe). AMSR2/GCOM-W1 surface soil moisture (LPRM) L3 1 day 25 km x 25 km descending V001 (2014).

[38] de Jeu, R. TMI/TRMM surface soil moisture (LPRM) L3 1 day 25 km x 25 km nighttime V001 (2013). URL https://disc.gsfc.nasa.gov/datacollection/LPRM_TMI_NT_SOILM3_001.html.

[39] CATDS. CATDS-PDC L3SM Aggregated - 3-day, 10-day and monthly global map of soil moisture values from SMOS satellite (2016). URL http://sextant.ifremer.fr/record/b57e0d3d-e6e4-4615-b2ba-6feb7166e0e6/.

[40] Wild, B. *et al.* VODCA2GPP – a new, global, long-term (1988–2020) gross primary production dataset from microwave remote sensing. *Earth System Science Data* **14**, 1063–1085 (2022). URL https://essd.copernicus.org/articles/14/1063/2022/. Publisher: Copernicus GmbH.

[41] Tao, S. *et al.* Increasing and widespread vulnerability of intact tropical rainforests to repeated droughts. *Proceedings of the National Academy of Sciences* **119**, e2116626119 (2022). URL https://www.pnas.org/doi/abs/10.1073/pnas.2116626119. Publisher: Proceedings of the National Academy of Sciences.

[42] Zhang, Y., Joiner, J., Alemohammad, S. H., Zhou, S. & Gentine, P. A global spatially contiguous solar-induced fluorescence (CSIF) dataset using neural networks. *Biogeosciences* **15**, 5779–5800 (2018). URL https://bg.copernicus.org/articles/15/5779/2018/.

[43] Skulovich, O., Li, X., Wigneron, J.-P. & Gentine, P. Global L-band equivalent AI-based vegetation optical depth dataset. *Scientific Data* **11**, 936 (2024). URL https://www.nature.com/articles/s41597-024-03810-2.

[44] Rahimi Tabar, M. R. *Analysis and Data-Based Reconstruction of Complex Nonlinear Dynamical Systems: Using the Methods of Stochastic Processes* Understanding Complex Systems (Springer International Publishing, Cham, 2019). URL http://link.springer.com/10.1007/978-3-030-18472-8.

[45] Shapiro, S. S. & Wilk, M. B. An Analysis of Variance Test for Normality (Complete Samples). *Biometrika* **52**, 591 (1965). URL https://www.jstor.org/stable/2333709?origin=crossref.

[46] Greene, W. H. *Econometric analysis* 5th ed edn (Prentice Hall, Upper Saddle River, N.J, 2003).

[47] Boettner, C. & Boers, N. Critical slowing down in dynamical systems driven by nonstationary correlated noise. *Physical Review Research* **4**, 013230 (2022). URL https://link.aps.org/doi/10.1103/PhysRevResearch.4.013230.

[48] Ben-Yami, M., Skiba, V., Bathiany, S. & Boers, N. Uncertainties in critical slowing down indicators of observation-based fingerprints of the Atlantic Overturning Circulation. *Nature Communications* **14**, 8344 (2023). URL https://www.nature.com/articles/s41467-023-44046-9.

[49] Friedl, M. & Sulla-Menashe, D. MODIS/Terra+Aqua Land Cover Type Yearly L3 Global 0.05Deg CMG V061 (2022). URL https://lpdaac.usgs.gov/products/mcd12c1v061/.

[50] Hansen, W. D. *et al.* Global forests are influenced by the legacies of past inter-annual temperature variability. *Environmental Research: Ecology* **1**, 011001 (2022). URL https://dx.doi.org/10.1088/2752-664X/ac6e4a. Publisher: IOP Publishing.

Online Methods

An overview of the study area, the grid cell selection to exclude human influence, and the monthly VPs analysed is shown in Fig. 1. For a grid cell to be included in our analysis, the HLU (*Croplands, Urban and Built-up Lands, and Cropland/Natural Vegetation Mosaics*) in the MODIS LCT data had to be 0 in every year between 2000 - 2021. Furthermore, the percentage of Evergreen Broadleaf Forests of a grid cell had to be at least 80 % in each year, and the forest loss according to the Hansen forest loss dataset must not accumulate to more than 10 % of the grid cell. Due to the limited data availability, we could not ensure the exclusion of grid cells influenced by HLU occuring after 2021. Thus, our results are based on the years 2000 - 2021. In Fig. 6, results are shown until November 2024 for VPs that are available, however, these results must be taken with a grain of salt due to the potential influence of HLU.

For this study, we compare four optical VPs, namely the Normalized Difference Vegetation Index (NDVI) from the Advanced Very High Resolution Radiometer (AVHRR) and the Visible Infrared Imaging Radiometer Suite (VIIRS), i.e. $\text{NDVI}^{\text{o+}}_{\text{AVHRR+VIIRS}}$ [34]. Furthermore, the NDVI from the Moderate Resolution Imaging Spectroradiometer (MODIS), i.e. $\text{NDVI}^{\text{o}}_{\text{MODIS}}$, its kernel version of the NDVI, i.e $\text{kNDVI}^{\text{o}}_{\text{MODIS}}$, and the Enhanced Vegetation Index, i.e. $\text{EVI}^{\text{o}}_{\text{MODIS}}$ [35]. Moreover, we test several single-sensor Vegetation Optical Depths (VOD) data sets, i.e. $\text{X-VOD}_{\text{AMSR-E}}$ and $\text{C-VOD}_{\text{AMSR-E}}$ from the Advanced Microwave Scanning Radiometer for EOS [36], $\text{C1-VOD}_{\text{AMSR2}}$ and $\text{C2-VOD}_{\text{AMSR2}}$ from the Advanced Microwave Scanning Radiometer 2 [37], $\text{X-VOD}_{\text{TMI}}$ from the Tropical Rainfall Measuring Mission's Microwave Imager [38] and $\text{L-VOD}_{\text{SMOS}}$ from the Soil Moisture and Ocean Salinity [39]. In addition to these datasets, we include the following new products that combine several sensors by simple merging or by using machine learning approaches: C- and Ku-VOD from the Vegetation Optical Depth Climate Archive (VODCA), i.e. $\text{C-VOD}^{+}_{\text{Moesinger et al.}}$ and $\text{Ku-VOD}^{+}_{\text{Moesinger et al.}}$ [25], a Gross Primary Production dataset, $\text{GPP}^{+}_{\text{Wild et al.}}$ [40], C-/KU-backscatter (BS) from radar observations $\text{C-/Ku-BS}^{+}_{\text{Tao et al.}}$ [41], solar-induced chlorophyll fluorescence (SIF), $\text{SIF}^{+}_{\text{Zhang et al.}}$ [42], and $\text{L-VOD}^{+}_{\text{Skulovich et al.}}$ [43].

We aggregated the data into monthly averages. Note that we discard grid cells with more than 12 missing months, and linearly interpolate for less than that. According to the standards of CSD-based resilience analyses, we then also deseasoned and detrended the data using Seasonal-Trend decomposition using LOESS (STL) with the proposed default parametrization. All analyses were performed on these detrended and deseasoned monthly anomalies.

3.1 The Ornstein-Uhlenbeck process

Critical Slowing Down (CSD)-based resilience metrics rely on an underlying OU process as in Eq. 1. The two most common resilience metrics are AC1 and variance, which for an OU process are given as

$$AC(1) = e^{\lambda \Delta t} \quad \text{and} \quad Var = \frac{\sigma^2}{-2\lambda}. \tag{2}$$

Yet, it is not trivial why vegetation should follow an OU process instead of a more complex process as expressed by a general stochastic differential equation

$$dx = f(x,t)dt + g(x,t)\eta. \tag{3}$$

On the way from from Eq. 3 to Eq. 1 the OU hypothesis also imposes the following assumptions: (i) the recovery rate is independent of the state, i.e. $f(x) \equiv \lambda x$, (ii) the diffusion is independent of the state, i.e. $g(x) \equiv \sigma$, (iii) the driving noise process is Gaussian, i.e. $\eta \sim \mathcal{N}(0,1)$, (iv) the driving noise process is white, i.e. not autocorrelated, and (v) the diffusion is constant over time, i.e. $g(t) \equiv \sigma$. Fan et al. [27] already validated the equilibrium assumption for tropical forests, which implies linearisability, i.e. assumption i. However, for completeness and as its a fundamental assumption

for the entire study, we counterchecked these results with an approach differing from [27] in Sec. S1.1 for each remote sensing vegetation dataset. The hypotheses ii and iii-v were assessed in Sect. S1.2 and Sect. 2.1, respectively. All of these approaches are explained in the following.

3.2 Properties of vegetation data w.r.t. Ornstein-Uhlenbeck assumptions

To obtain estimates of the drift and diffusion in dependence on the state, the Kramers-Moyal expansion [28, 44] was employed. Namley, the first two Kramers-Moyal components refer to

$$\mathrm{KM1(x)} = f(x) \stackrel{!}{=} \lambda x \quad \text{and} \quad \mathrm{KM2(x)} = \frac{1}{2} g(x) \stackrel{!}{=} \frac{1}{2}\sigma^2, \tag{4}$$

derived from the mean and variance of kernel-based increment distributions around the given value x. Figs. S1.7 and S1.9 display these series. In the case of an OU process, KM1 is a polynomial of order 1 and KM2 a polynomial of order 0. Hence, polynomials of order 1 - 4 and 0 - 4 were fitted to KM1 and $\sqrt{2 \cdot KM2}$, respectively. Since all models differ in the degree of freedom, we measure the goodness of these fits with the Bayesian Information Criterion (BIC). The BIC penalizes higher degrees of freedom of different models. The results in Fig. S1.8 suggest that the higher-order polynomials do not enhance the fit sufficiently to warrant their complexity and thus, the vegetation data might be represented by state-space independent drift and diffusion.

The next assumptions mentioned above demand Gaussian and white noise. Due to the likely state invariance of the drift for all VPs, and the additivity of the noise, the time series of σdW_t can be obtained as the residuals of an Generalized Least Squares regression with AR covariance structure (GLSAR) of dx onto x, as suggested by Boers [20]. GLSAR allows regression in cases with autocorrelated noise. This results in noise time series for each single location, see Fig. S1.10. We tested if these noise time series followed a normal distribution, using the Shapiro-Wilk test [45]. This two-sided test's null hypothesis is that the data are normally distributed. In Fig. 2a we show this test's p-values.

We tested for serial correlation in the noise using the Breusch–Godfrey test. The test was performed with the default parametrization proposed by Greene [46] and defaults to maximum 10 lags for our data. However, results for other parametrizations are shown in Sect. S1.5. The Breusch–Godfrey test evaluates the null hypothesis of no autocorrelation against the alternative that autocorrelation is present as a one-sided test. P-values were calculated based on the chi-squared distribution of the test statistic and are displayed in Fig. 2b.

To investigate if the diffusion is constant over time, i.e. $g(t) \equiv \sigma$, noise time series were extracted via GLSAR on sliding windows of 5 years. For a given window t, the standard deviation of the regression's residual gives the $\sigma(t)$. We note that this approach does not take into account that the data are discretely sampled from a continuos process. Namely, if the underlying process follows $dx/dt = \lambda x + \sigma dW_t$, this implies that the samples follow $x_{t+1} = x_t \cdot e^{\lambda \cdot 1} + \eta$ with $\eta \sim \mathcal{N}(0, \tilde{\sigma})$. Hence,

no corrections are necessary to test if the noise is white and Gaussian as above, but to investigate the time evolution of σ, we must get its continuous version from the inversion of the formula that relates the two [47] (note that they define the restoring rate negatively with $dx = -\lambda x + \sigma dW_t$), which is

$$\tilde{\sigma}^2 = -\frac{1}{2\lambda}\sigma^2(1 - e^{2\lambda\Delta t}). \tag{5}$$

Then, polynomials of degree 0 - 4 were fitted to the obtained time series of $\sigma(t)$, as for the drift and diffusion, and their goodness-of-fit was assessed via the BIC. In addition, as a measure of the steadiness of change, the Kendall's τ of the amplitude time series was calculated.

3.3 Theoretical metrics of resilience

To investigate resilience changes, statistical metrics are usually employed to infer resilience in grid cells and on time windows that had not experienced significant perturbations. As mentioned before, the most common resilience metrics are AC1 and variance. Yet, several other metrics had been proposed in literature. In this study, we compared the performance of nine resilience metrics. These are

1. AC1,
2. variance,
3. λ_{AC1} inferred from AC1, i.e.

$$\lambda_{\mathrm{AC1}} = \frac{1}{\Delta t} log(AC1) \quad \text{with} \quad \Delta t = 1,$$

4. $\hat{\lambda}_{\mathrm{Variance}}$ inferred from metric 2, i.e. variance (Var) under the assumption of $\sigma \equiv 1$, i.e.

$$\hat{\lambda}_{\mathrm{Variance}} = -\frac{1}{2 \cdot Var},$$

5. $\lambda_{\mathrm{Variance}/\sigma(LR)^2}$ inferred from variance with $\hat{\sigma}$ estimated from the residual of an ordinary least squares regression of dx onto x following Smith and Boers [15], i.e.

$$\lambda_{\mathrm{Variance}/\sigma(LR)^2} = \frac{1}{2\Delta t} log\left(1 - \frac{\sigma^2}{Var}\right) \quad \text{with} \quad \Delta t = 1,$$

6. $\hat{\lambda}_{\mathrm{GLSAR}}$ as the regression coefficient of a Generalized Least Squares regression with AR covariance structure (GLSAR) of dx onto x, as suggested by Boers [20].
7. λ_{GLSAR} inferred from the regression coefficient $\hat{\lambda}_{\mathrm{GLSAR}}$ as explained by Ben-Yami et al. [48]

$$\lambda_{\mathrm{GLSAR}} = log(1 + \hat{\lambda}_{\mathrm{GLSAR}}),$$

8. λ_{KM} calculated as the slope of the first Kramers-Moyal component $KM1(x)$, following Morr et al. [28], and

9. spatial correlation as the mean temporal correlation of the centroid cell with its neighbouring grid cells that lay within a 100 km radius, following Blaschke et al. [18].

Note that we have defined all metrics such that higher values relate to lower resilience, and that the λ, on the other hand, denotes the 'continuous' version as in Eq. (1). The $\hat{\lambda}$ denotes the 'discrete' version, which results from inversion of the Eqs. 2 without correcting for the fact that the statistics come from discrete sampling of the continuous data. This is used in much of the literature.

3.4 Empirical measure of resilience

The above mentioned metrics are statistical estimations of resilience. Resilience is strictly defined as the recovery rate of a system when recovering from perturbations. As such, whenever there is a large and singular perturbation, resilience can be estimated by fitting an exponential function to the recovery period. In general, it is not obvious that the theoretical (diagnostic) metrics of resilience discussed above capture the recovery rate from significantly large perturbations. Following Smith et al. [9], we detected significant perturbations and computed an empirical resilience estimate $\lambda_{\text{empirical}}$ by fitting an exponential function to the recovery period.

To identify perturbation times in a time series, we applied a sliding-window approach of 9 months. As a coarse estimate of the local derivate, we computed the difference between the mean of the preceding and following half-window of each time point. Noise in this signal was smoothed with a first-order Savitzky–Golay filter (7 points), which allows preserving peak structures. Of these peaks, only the most pronounced changes in the 99th percentile were taken into account. The following 8 months were defined as the perturbation period from which the minimum is taken as the final perturbation time. For each transition date, the local minimum within the post-transition segment of three years was identified, and the subsequent trajectory was used for fitting. To account for baseline shifts, the time series was re-centred by subtracting its approximate equilibrium level (estimated as the mean of the latter half of the segment). An exponential function of the form $a + e^{\lambda t}$ was fitted under the conditions $a \leq 0$ and $\lambda \leq 0$. For each fit, the coefficient of determination R^2 was calculated to assess the goodness of the fit. For the λ to be taken into account, the following additional criteria had to be fulfilled. First, the drop in the raw time series from the beginning of the detected perturbation period (8 months) to its actual perturbation time must have dropped below 10 % of the entire time series' standard deviation (to account for different variance in the different data sets). Second, the p-value of a two-sided Kolmogorov-Smirnov test, comparing the pre- and post-perturbation 3 years' raw data mean and variance, must be below 0.05 to rule out recoveries to *new* equilibria. Third, the R^2 must be highest compared to all other fits *in the same time series*. Fourth, we exclude unphysical recovery rates by requiring $\lambda > -3$. Compared to Smith et al. [9], we employ their 'method 1' adapted but changed the following. First, we kept the 9-months-window for perturbation detection, which are then 9 instead of 18 data points as their data is bi-monthly. Second, we approximated the equilibrium level after the perturbation as the mean of the latter half of the recovery period, instead of

the entire one, to avoid overly good evaluation of fits in cases of extremely fast recovery. Third, we define the necessary drop in the raw time series in relation to the time series' standard deviation (instead of a fixed value) to account for different variation in the different data sets. Fourth, in the rare case of several perturbations within one single time series, instead of taking the mean, we use the one with the best R^2.

For different minimal levels of R^2 of $\lambda_{\text{empirical}}$, the correlation of $\lambda_{\text{empirical}}$ with the theoretical resilience metrics was measured in terms of Pearson correlation across space. Results were only taken into account if at least ten grid cells showed $\lambda_{\text{empirical}}$ with minimum R^2. Otherwise, the results were omitted. To assess the robustness of the results, the Pearson correlation was compared to the Spearman's rank correlation (see Figs. S2.17 and S2.18). The significance of the Pearson correlation coefficients was assessed using a two-sided t-test with n-2 degrees of freedom, testing the null hypothesis that the true correlation is zero. The significance level is $\alpha = 0.025$. Note that we do not correct the p-values for potential correlations between the grid cells, assuming that grid cells with detected significant perturbations are sparsely scattered, compare the maps in Fig.S2.15.

3.5 Resilience analysis

To assess resilience changes in tropical forests we calculated the chosen resilience metric on sliding windows of 5 years. The change was then assessed by linear trends (discussed in the main text) and by Kendall's τ (discussed in Sect. S2.4).

Data Availability. For the exclusion of anthropogenic influence in the study area, the MODIS land cover type (LCT) dataset[49] (last accessed 21/12/2024 at https://e4ftl01.cr.usgs.gov/MOTA/MCD12C1.061/) and the Hansen forest loss dataset[50] (last accessed 31/05/2022 at https://glad.earthengine.app/view/global-forest-change) were used.

The following lists the vegetation data, the access date, and the download link:

- $\text{NDVI}^{\text{o}}_{\text{MODIS}}$ and $\text{EVI}^{\text{o}}_{\text{MODIS}}$ from Didan [35]: last accessed 21/12/2024 at https://e4ftl01.cr.usgs.gov/MOLT/MOD13C2.061/
- $\text{NDVI}^{\text{o+}}_{\text{AVHRR+VIIRS}}$ from Vermote and NOAA CDR Program [34]: last accessed 21/12/2024 at https://www.ncei.noaa.gov/data/land-normalized-difference-vegetation-index/access/
- C-/Ku-$\text{BS}^{+}_{\text{Tao et al.}}$ from Tao et al. [41]: last accessed 16/02/2025 at https://figshare.com/articles/dataset/Radar_Rainforest/14061428/6
- $\text{SIF}^{+}_{\text{Zhang et al.}}$ from Zhang et al. [42]: last accessed 27/09/2024 at https://figshare.com/articles/dataset/CSIF/6387494 ('clear.daily')
- L-$\text{VOD}^{+}_{\text{Skulovich et al.}}$ from Skulovich et al. [43]: last accessed 03/09/2024 at https://zenodo.org/records/10306095
- C-$\text{VOD}^{+}_{\text{Moesinger et al.}}$ and Ku-$\text{VOD}^{+}_{\text{Moesinger et al.}}$ from Moesinger et al. [25]: last accessed 07/10/2021 at https://zenodo.org/records/2575599#.YLdOwZNKhB4
- $\text{GPP}^{+}_{\text{Wild et al.}}$ from Wild et al. [40] last accessed 24/07/2023 at https://researchdata.tuwien.ac.at/records/1k7aj-bdz35
- X-VOD_{TMI}: from de Jeu [38]: last accessed 01/01/2024 at https://hydro1.gesdisc.eosdis.nasa.gov/data/WAOB/LPRM_TMI_NT_SOILM3.001/
- X-$\text{VOD}_{\text{AMSR-E}}$ and C-$\text{VOD}_{\text{AMSR-E}}$ from Vrije Universiteit Amsterdam (Richard de Jeu) and NASA GSFC (Manfred Owe) [36]: last accessed 08/11/2022 at https://hydro1.gesdisc.eosdis.nasa.gov/data/WAOB/LPRM_AMSRE_D_SOILM3.002
- C1-$\text{VOD}_{\text{AMSR2}}$ and C2-$\text{VOD}_{\text{AMSR2}}$ from Vrije Universiteit Amsterdam (Richard de Jeu) and NASA GSFC (Manfred Owe) [37]: last accessed 21/12/2024 at https://hydro1.gesdisc.eosdis.nasa.gov/data/WAOB/LPRM_AMSR2_D_SOILM3.001/
- L-VOD_{SMOS} was obtained from the "Centre Aval de Traitement des Données SMOS" (CATDS), operated for the "Centre National d'Etudes Spatiales" (CNES, France) by IFREMER (Brest, France) [39] and last accessed 21/12/2024 at https://data.catds.fr/cpdc/Land_products/GRIDDED/L3SM/RE07/MIR_CLF3MA/

Code availability. All Python code used for the analysis is available from the author upon request (lblaschke@pik-potsdam.de) or on GitHub at ??.

Acknowledgements. The authors sincerely thank Fabian Telschow for inspiring this project. This study received support from the European Space Agency Climate Change Initiative (ESA-CCI) Tipping Elements SIRENE project (contract no. 4000146954/24/I-LR). This is ClimTip contribution #X; the ClimTip project has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement No. 101137601. MH was supported by the Serrapilheira Institute under the grant number Serra-1809-18983. LB was supported by a fellowship from the German Academic Exchange Service (DAAD). NB acknowledges

funding by the Volkswagen foundation. The authors gratefully acknowledge the Ministry of Research, Science and Culture (MWFK) of Land Brandenburg for supporting this project by providing resources on the high performance computer system at the Potsdam Institute for Climate Impact Research.

Competing interests. The authors declare no competing interests.

Supplementary information. This publication is accompanied by supplementary information.

Supplementary Information for

'Resilience Loss of Tropical Forests in recent decades'

Lana L. Blaschke[1,2*], Sebastian Bathiany[1,2], Marina Hirota[3], Niklas Boers[1,2,4]

[1] Munich Climate Center and Earth System Modelling Group, Department of Aerospace and Geodesy, TUM School of Engineering and Design, Technical University of Munich, Lise-Meitner-Straße 9, Ottobrunn, 85521, Bavaria, Germany

[2] Complexity Science, Potsdam Institute for Climate Impact Research, Telegrafenberg A 31, Potsdam, 14473, Brandenburg, Germany.

[3] Department of Physics, Federal University of Santa Catarina, Florian´opolis, Brazil.

[4] Department of Mathematics and Global Systems Institute, University of Exeter, Exeter, UK.

S1 Validation of Ornstein-Uhlenbeck assumptions in the data

S1.1 Assumption i: The drift is independent of the state

As a first step, we assessed the assumption of the drift's independence of the state space (Methods). The estimates in dependence of the state variable shown in Fig. S1.7 suggest a linear relationship between x and the drift. Fig.S1.8 compares the distribution of BICs for different polynomial degrees for each VP. Here, the violet violins refer to the results for the drift. It shows that, while higher polynomial orders exhibit slightly lower BICs, the distributions do not allow to rule out that the first-order polynomial is a good fit for the drift. Thus, we might assume state space independence of the drift in any of the datasets.

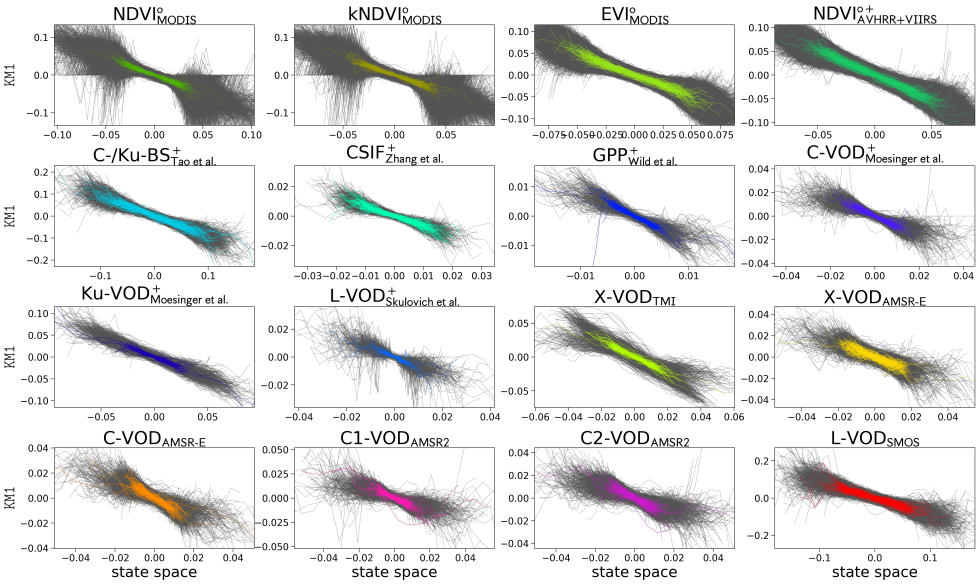

Fig. S1.7 The first Kramers-Moyal coefficient in dependence of the state variable. Some random grid cells are highlighted for visibility. As the first Kramers-Moyal coefficient resembles the drift term, which in an Ornstein-Uhlenbeck process would be λx, we expect a straight line.

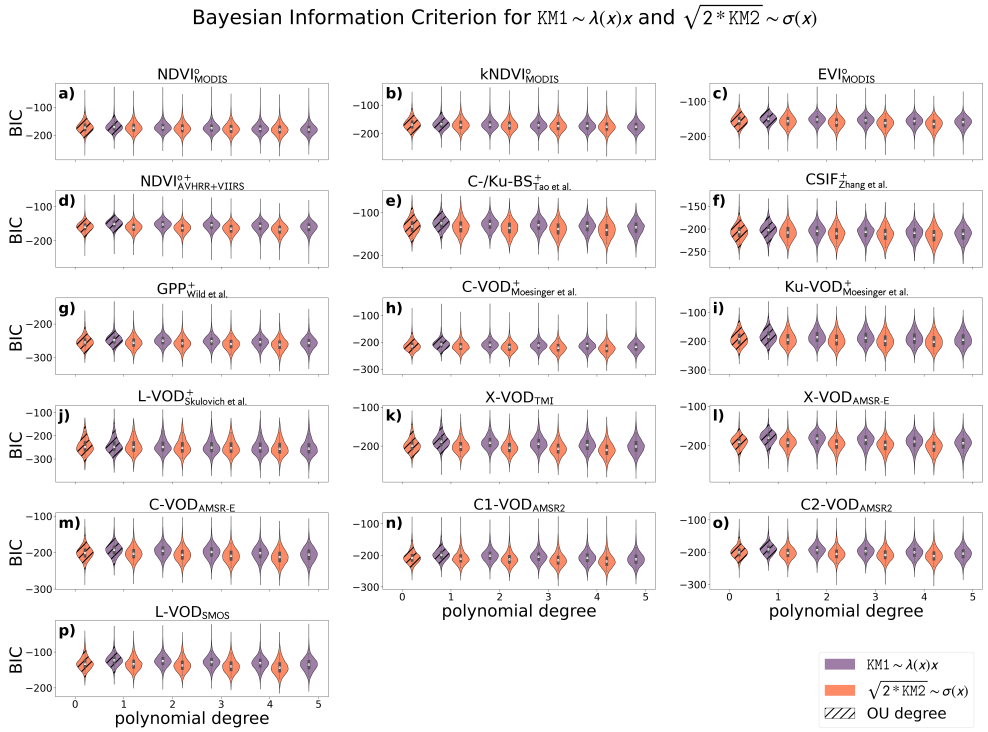

Fig. S1.8 Distribution of the BIC for polynomials fitted to the estimates of drift (violet) and diffusion (orange). For each grid cell, they are estimated in dependence of the state space. Polynomials of order 1 to 5 (drift) and 0 to 4 (diffusion) are fitted to the estimates, where 1 and 0 would be the expected best fits according to the OU theory (marked as hatched). This Figure shows each VP's distribution of the Bayesian Information Criterion (BIC) across space for each polynomial degree.

S1.2 Assumption ii: The noise is additive

The same increment-based method allows to estimate the amplitude of the diffusion term, $\sigma(x)$. In Fig. S1.9 showing the series in dependence of x no strong dependence of x can be seen. More concisely, a zero-order polynomial must be sufficient to fit the diffusion estimate. In Fig. S1.8 the orange violins mark the results for the diffusion. As for the drift, the BICs for higher degrees of freedom are slightly lower, yet they do not disprove that the zero-order polynomial is suitable to approximate the diffusion term. This implies that we might assume additivity of the noise for all VPs.

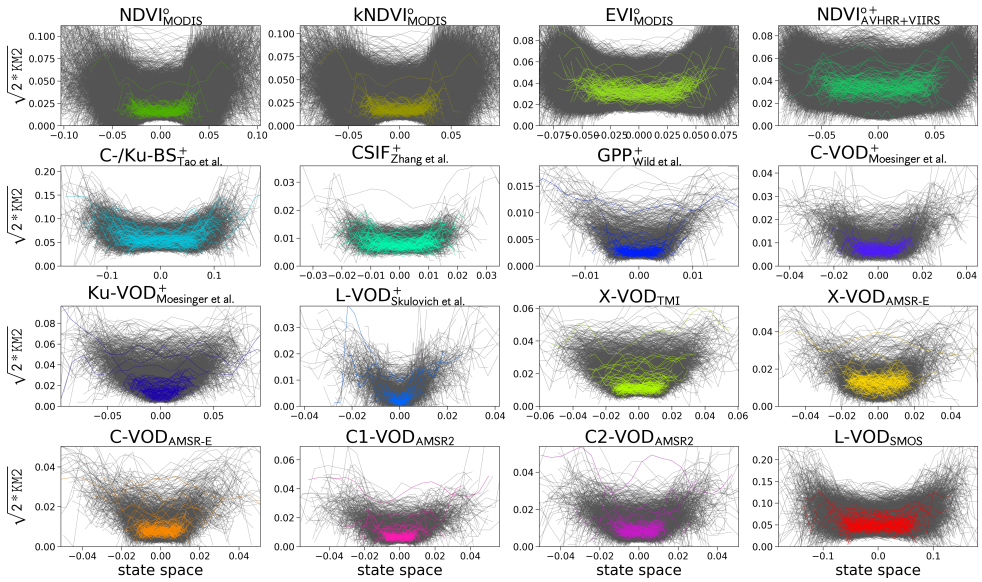

Fig. S1.9 The second Kramers-Moyal coefficient in dependence of the state variable. Some random grid cells are highlighted for visibility. As the second Kramers-Moyal coefficient resembles the diffusion term, which in an Ornstein-Uhlenbeck process would be σdW_t, we expect this Coefficient to be independent of the state variable and hence a horizontal line.

S1.3 Retrieval of noise time series

For the investigation of the noise properties, noise time series were retrieved for each single VP and each grid cell thereof, as the residual of a GLSAR. This results in noise time series for each grid cell, some of them examplarily depicted in Fig. S1.10.

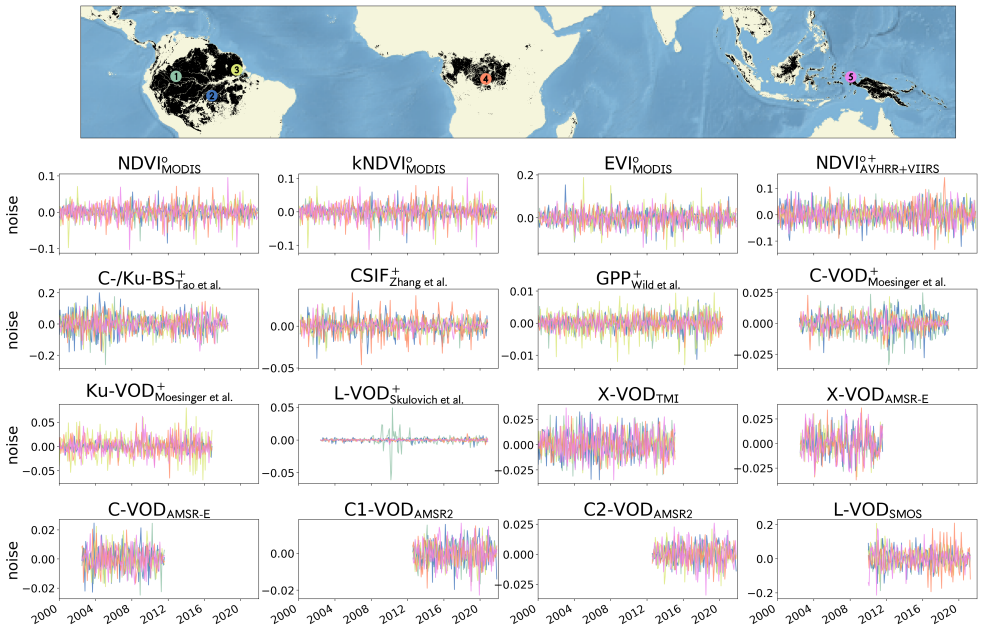

Fig. S1.10 Exemplary noise time series as derived by GLSAR of dx onto x.

S1.4 Assumption iii: The noise is Gaussian

The Shapiro-Wilk test was applied to the noise time series to test for the noise being Gaussian, as assumed by the Ornstein-Uhlenbeck process. Fig. S1.11 shows that the results are robust w.r.t. to different parametrizations of GLSAR for retrieving the noise.

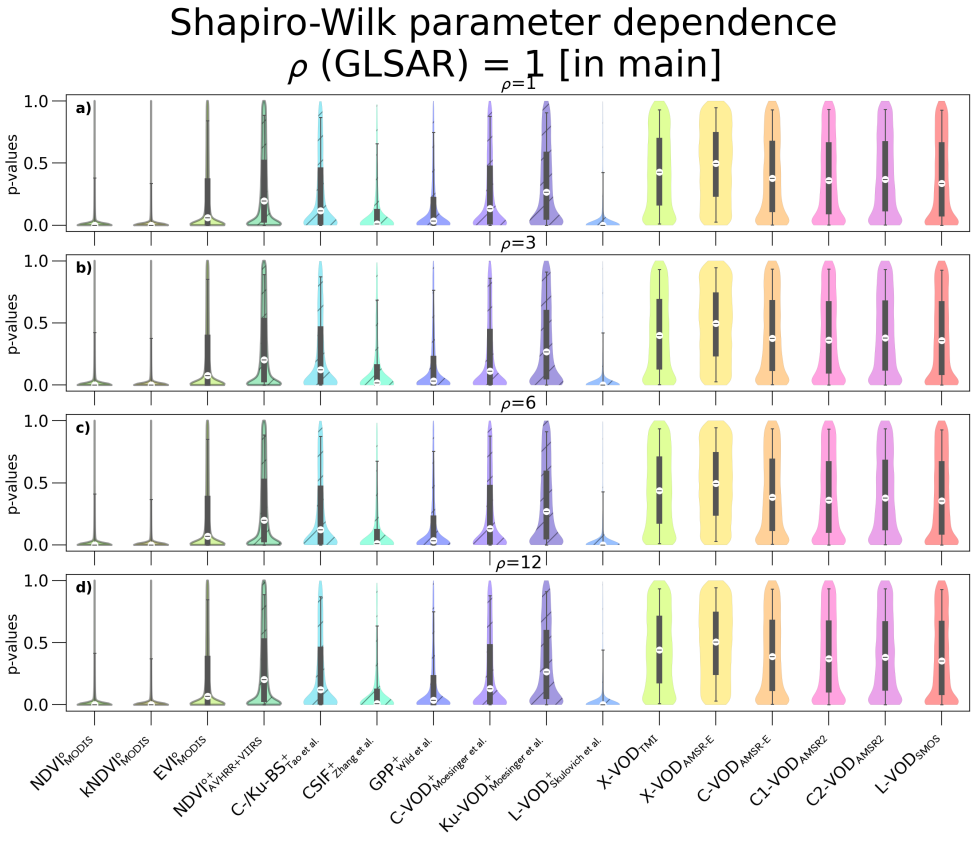

Fig. S1.11 Shapiro-Wilk p-values for different parametrizations of GLSAR to derive the noise time series as regression of dx onto x.

S1.5 Assumption iv: The noise is white

Likewise as for Gaussianity, the Breusch-Godfrey can be applied to the same noise time series. This test assumes whiteness, i.e. no autocorrelation over time, as its null-hypothesis. Fig. S1.12 shows that the results are varying w.r.t the parametrization of GLSAR for retrieving the noise (ρ) as well as the number of lags considered in the Breusch-Godfrey test itself (nlags). Yet, it can be seen that in any case, the majority cells for any dataset shows extremely low p-values, indicating that many cells have an autocorrelated noise structure.

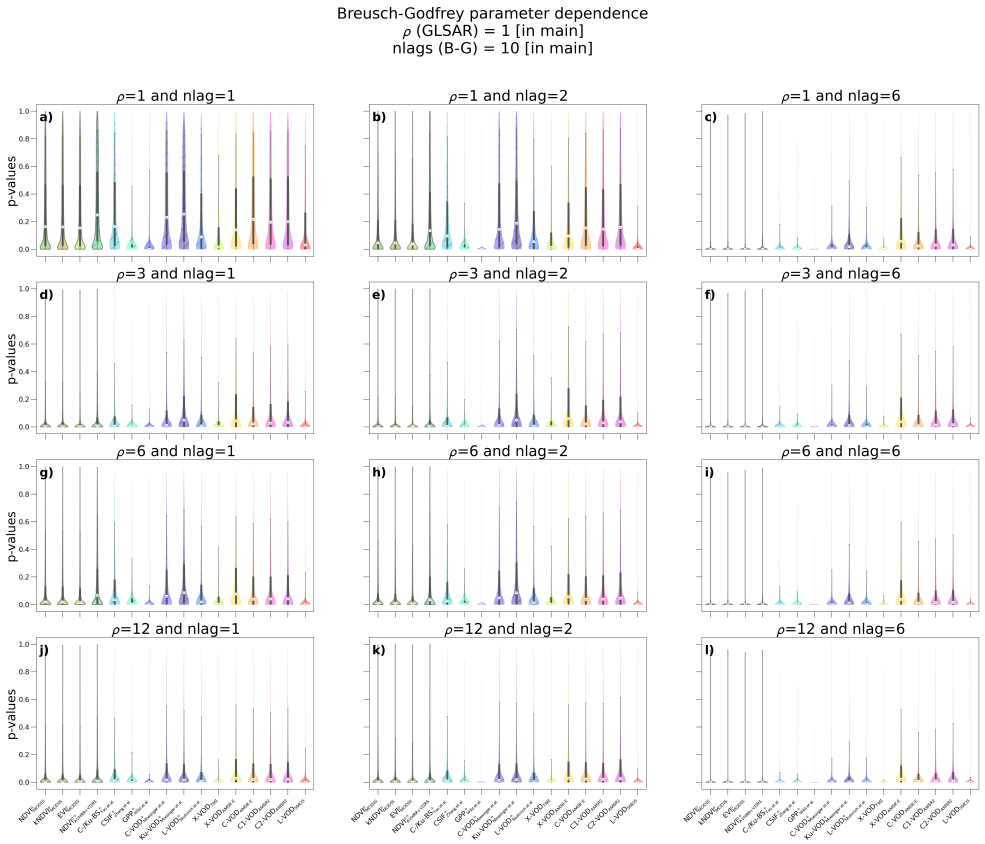

Fig. S1.12 Breusch-Godfrey p-values for different parametrizations of GLSAR to derive the noise time series as regression of dx onto x.

S1.6 Assumption v: The amplitude of the noise is constant over time

The last important assumption when mathematically deriving AC1 and variance as metrics of λ is the assumption of a constant noise amplitude σ. Hence, Fig. S1.13a-p show the distributions of the BIC of polynomials of order 0-4 fitted to $\sigma(t)$ (Methods).

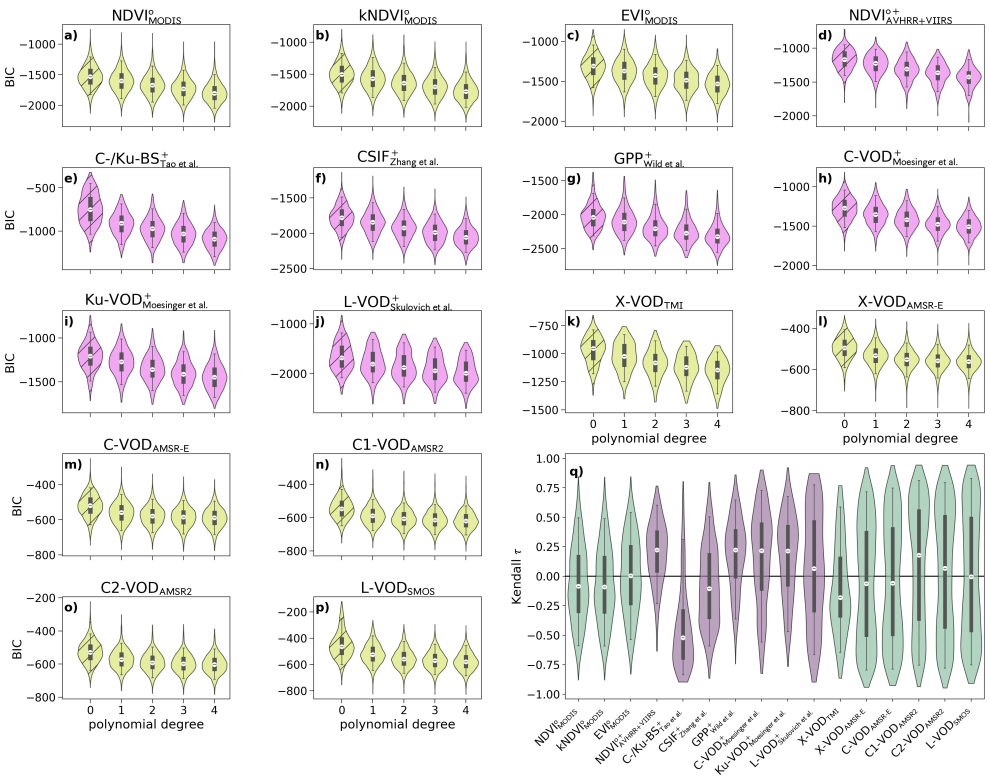

Fig. S1.13 Assessment of the noise level $\sigma(t)$ over time. Panels **a-p)** depict, for the different datasets, the distribution of BIC for polynomials of degree 0 to 4 fitted to the estimates of $\sigma(t)$ calculated on sliding windows of 5 years. Constant noise would be indicated by smallest BICs for polynomials of order 0 (hatched). The Kendall τ in panel **q)** is the same as in Fig. 2. It shows that a constant $\sigma(t)$ cannot be assumed, in particular in cases of merged data (pink/violet).

S2 Comparison to empirical recovery rates

Fig. S2.14 shows an example of a vegetation time series that experienced a large perturbation towards the end of the year 2017 and recovered afterwards. The ecological recovery rate is then estimated as the exponent λ of an exponential function $e^{\lambda t} + c$, with c being a constant, fitted to the time series' following three years starting from the most perturbed state for $t = 0$. How well the exponential fitted the vegetation recovery is assessed by the coefficient of determination R^2. Fig. S2.15 shows maps of $\lambda_{\text{empirical}}$ and the corresponding R^2.

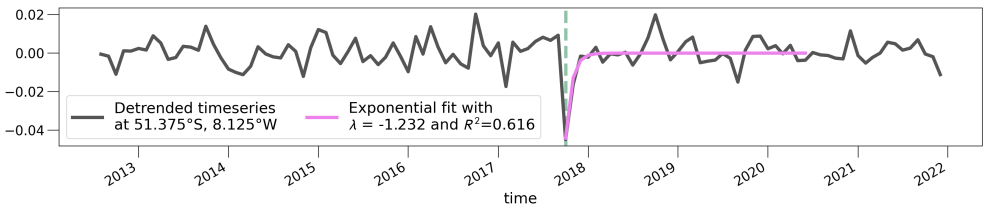

Fig. S2.14 Exemplary time series from C1-VOD$_{\text{AMSR2}}$ that exhibited a large perturbation in the end of the year 2017 (green vertical line), followed by a recovery. The pink line shows the exponential fit e^{λ} to this recovery. The exponent λ is the estimated ecological recovery rate and the R^2 quantifies its goodness-of-fit.

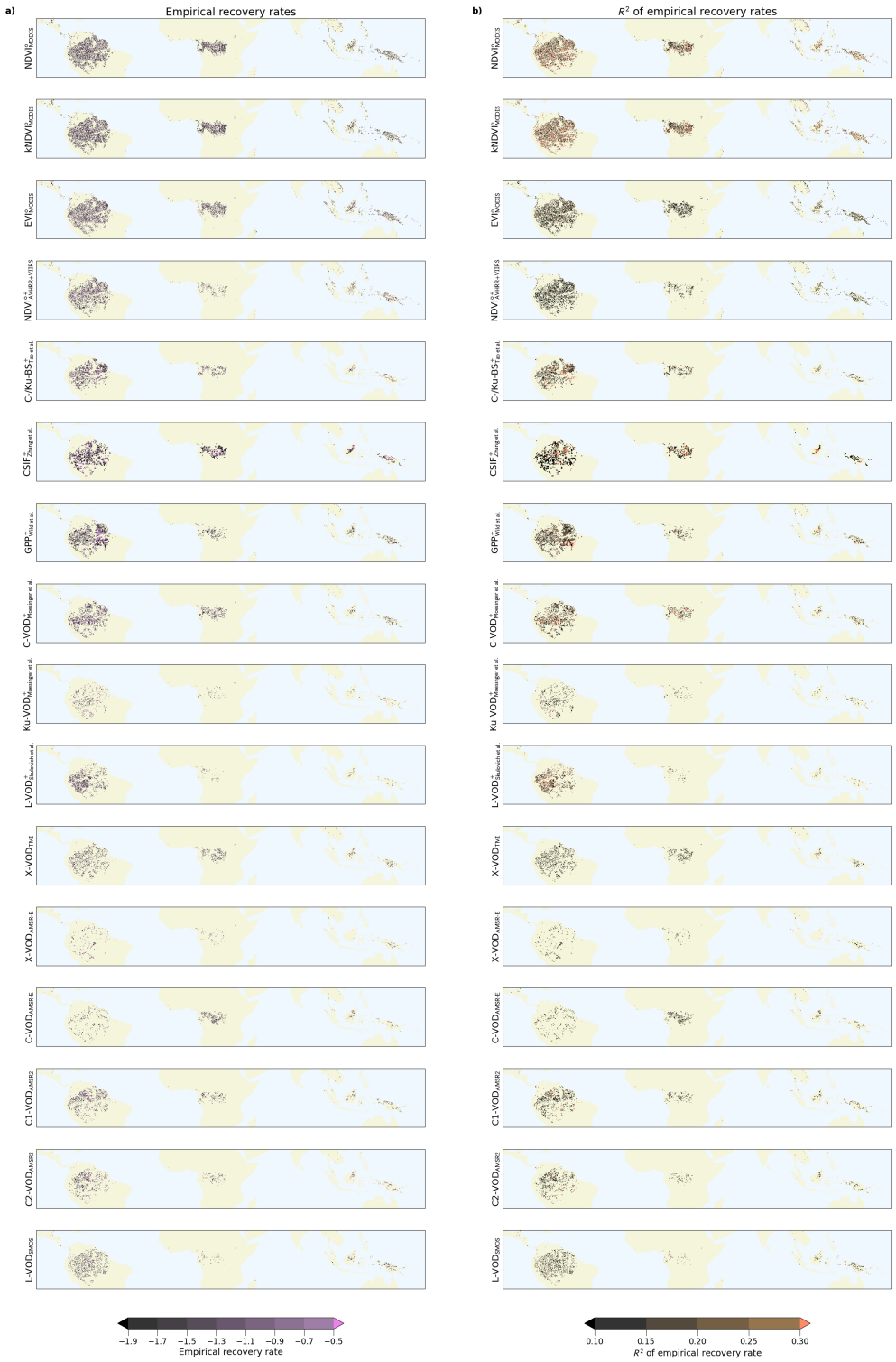

Fig. S2.15 Maps of a) the empirical recovery rate and b) the corresponding goodness-of-fit level R^2 as detected for each VP.

S2.1 Results of the correlation with the empirical recovery rates for all theoretical metrics

To keep the main text clean, in Fig. 3b the single VP's Pearson correlations of $\lambda_{\text{empirical}}$ are shown only with $\hat{\lambda}_{\text{GLSAR}}$. In Fig. S2.16, the correlation for the other theoretical metrics is shown.

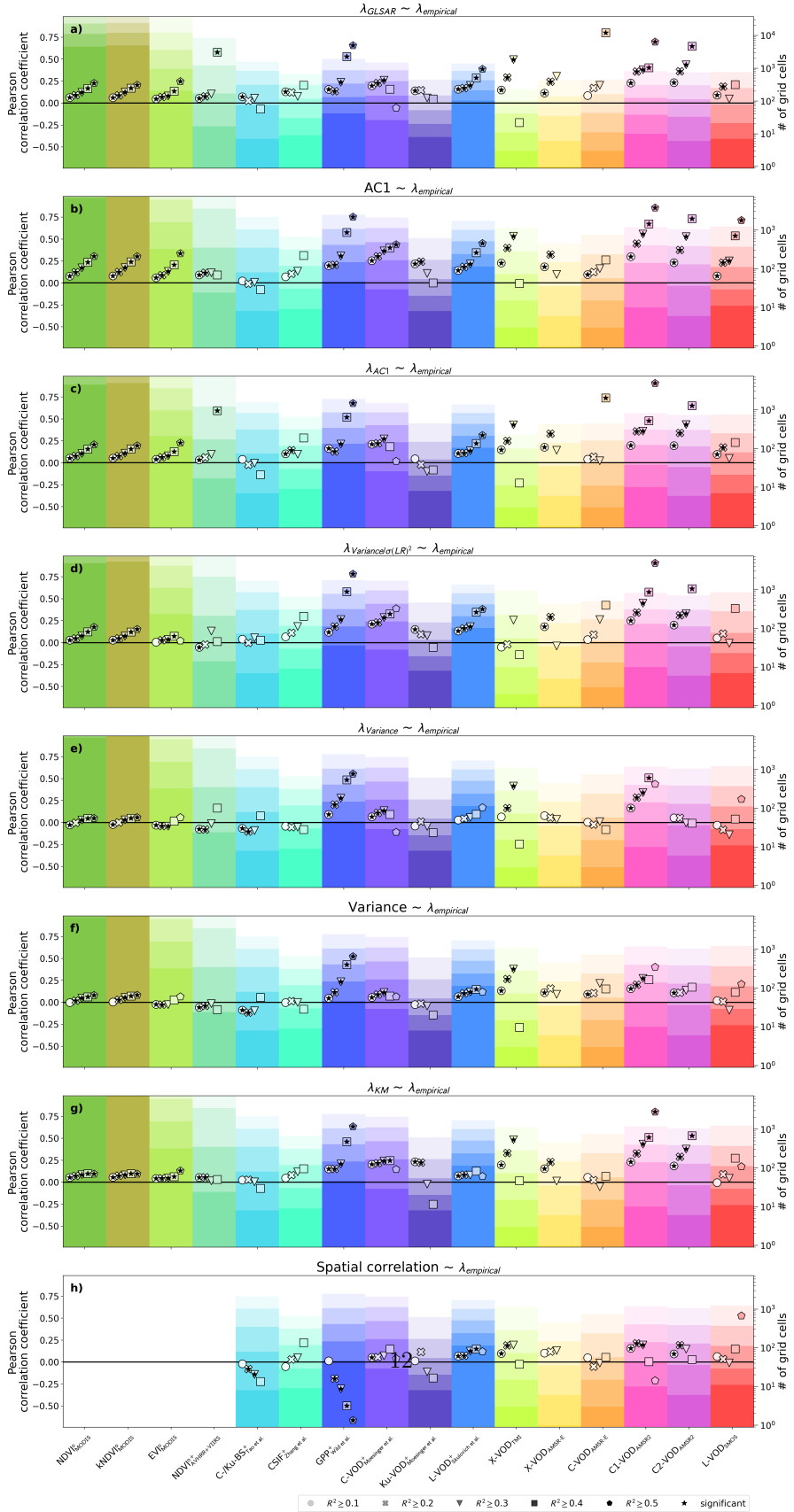

Fig. S2.16 The empirical recovery rate's correlation with its theoretical estimates per VP for the remaining theoretical estimates, i.e. a) λ_{GLSAR}, b) AC1, c) λ_{AC1}, d) $\lambda_{\text{Variance}/\sigma(LR)^2}$, e) $\hat{\lambda}_{\text{Variance}}$, f) variance, g) λ_{KM}, and h) spatial correlation, respectively. Compare to Fig. 3b which depicts the same results for the final metric $\hat{\lambda}_{\text{GLSAR}}$.

S2.2 Results of the correlation with the empirical recovery rates based on Spearman's correlation

We suggest that the Pearson correlation coefficient is more suitable in our case, as our main goal is to estimate the proximity of the theoretical λ-estimates to $\lambda_{\text{empirical}}$. Yet, it is important to see that the results are robust when considering Spearman's rank correlation instead, compare Figs. 3 and S2.17 for the comparison averaged over VPs (a) as well as the results for comparing $\hat{\lambda}_{\text{GLSAR}}$ to $\lambda_{\text{empirical}}$ (b). In addition, for the VP-specific Pearson and Spearman correlation coefficients are depicted in Figs. S2.16 and S2.18, respectively, for all other theoretical resilience metrics.

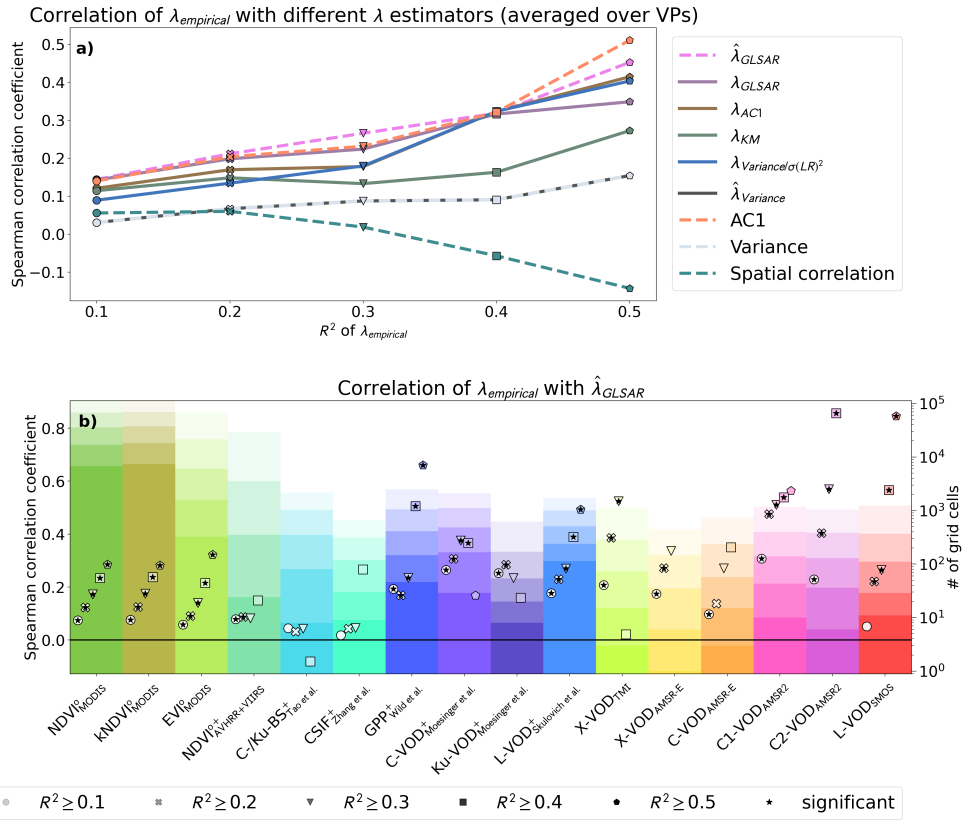

Fig. S2.17 Spearman correlation of the theoretical resilience metrics with the $\lambda_{\text{empirical}}$, a) averaged over all VPs, in dependence of the goodness-of-fit of the $\lambda_{\text{empirical}}$. Panel b) shows the Pearson correlation of $\hat{\lambda}_{\text{GLSAR}}$ separately for each VP. The different markers encode different goodness-of-fit levels, indicated by R^2 . The amount of cells for which a $\lambda_{\text{empirical}}$ of at least the given r2 , as well as the theoretical metric, could be calculated, is indicated by the histograms, shaded as the corresponding marker. Note their log-scaled y-axis on the right-hand side. Compare to Fig. 3.

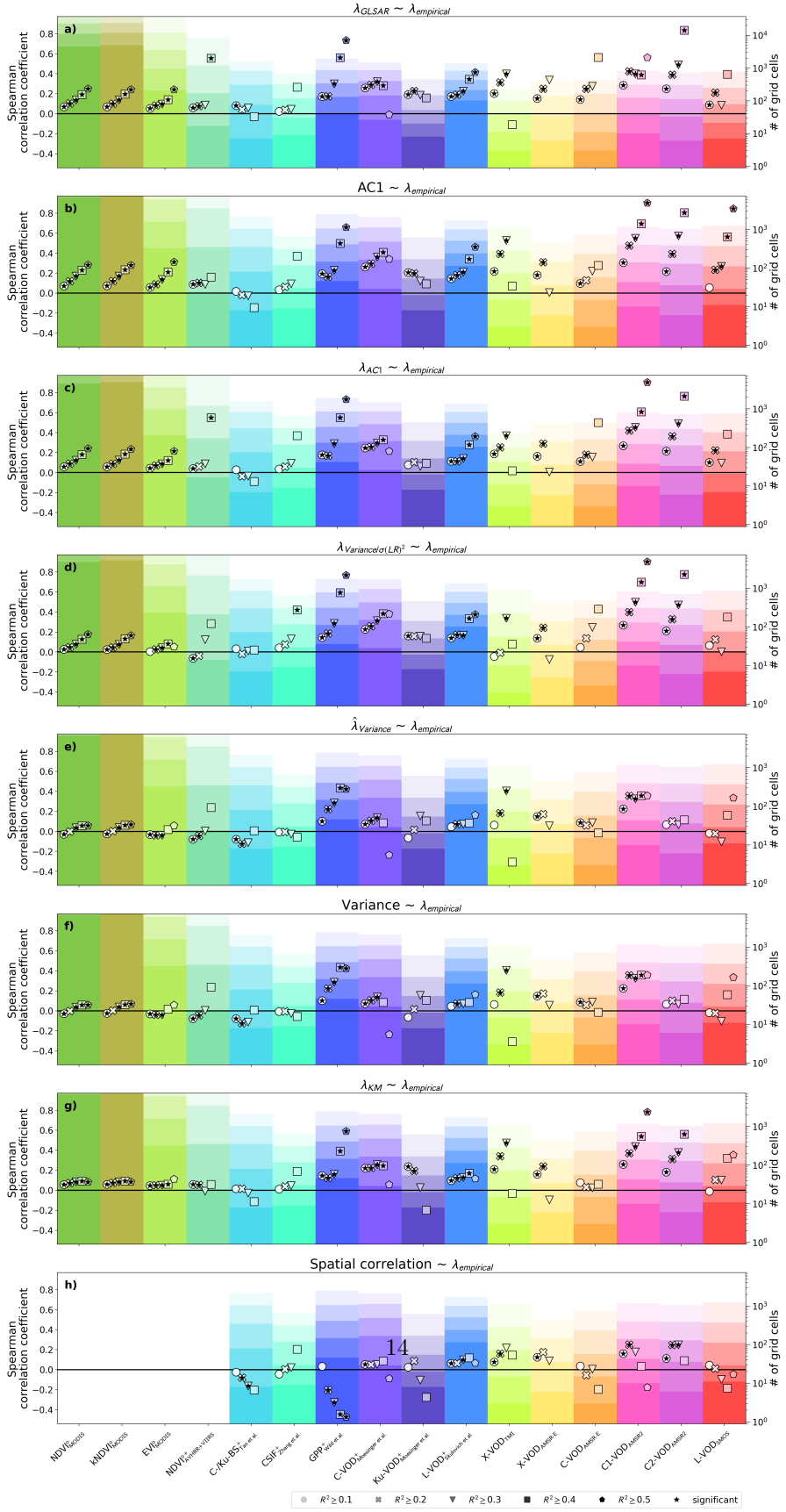

Fig. S2.18 Spearman correlation of the empirical recovery rate and its theoretical estimates for λ_{GLSAR}, AC1, λ_{AC1}, $\lambda_{\mathrm{Variance}/\sigma(LR)^2}$, $\hat{\lambda}_{\mathrm{Variance}}$, variance, λ_{KM}, and spatial correlation, respectively. Compare to Fig. S2.17b as well as S2.16.

S2.3 Problems in X-VOD$_{\text{AMSR-E}}$

Fig. S2.19 shows the region-wise average of $\hat{\lambda}_{\text{GLSAR}}$ for X-VOD$_{\text{AMSR-E}}$. All three regions show an extremely similar pattern in $\hat{\lambda}_{\text{GLSAR}}$-change. For this reason, we suspect that the signal $\hat{\lambda}_{\text{GLSAR}}$ picks up in X-VOD$_{\text{AMSR-E}}$ is rather a symptom of measurement errors than of the vegetation. Hence, we exclude it from our resilience analysis.

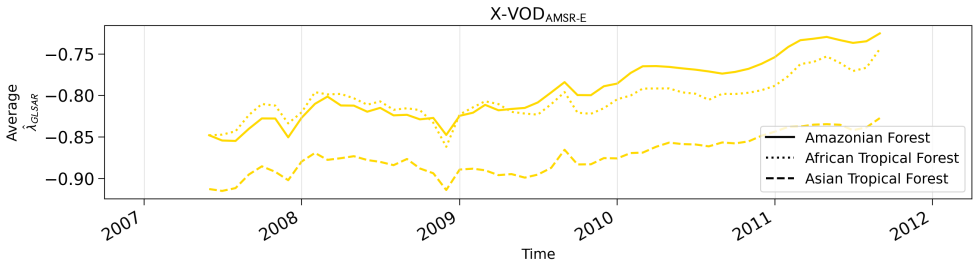

Fig. S2.19 Average $\hat{\lambda}_{\text{GLSAR}}$ time series of X-VOD$_{\text{AMSR-E}}$ for the Amazonian Forest (solid line), the African Tropical Forest (dotted line), and the Asian Tropical Forest (dashed line).

S2.4 Robustness of the resilience analysis

In the main text, we assess the change of the resilience metric by linear trends. Yet, Figs. S2.20-S2.25 show that these changes are robust w.r.t. Kendall's τ, indicating that increases are detected as a linear trend are usually also increasing steadily.

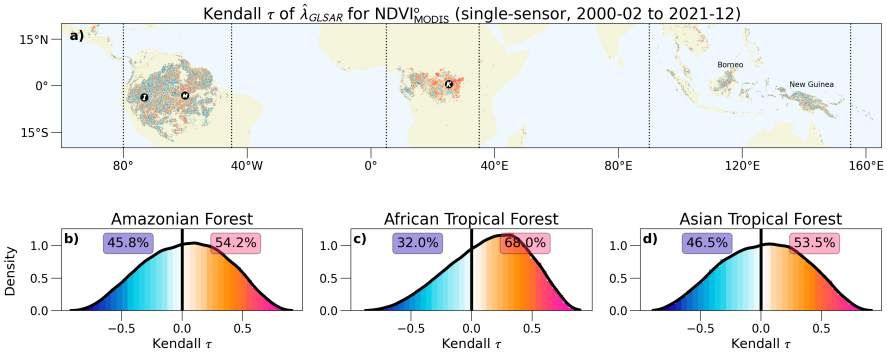

Fig. S2.20 Resilience change accessed as Kendall τ of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for NDVI°$_{\mathrm{MODIS}}$. Compare to the linear trend in Fig. 5a-d. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

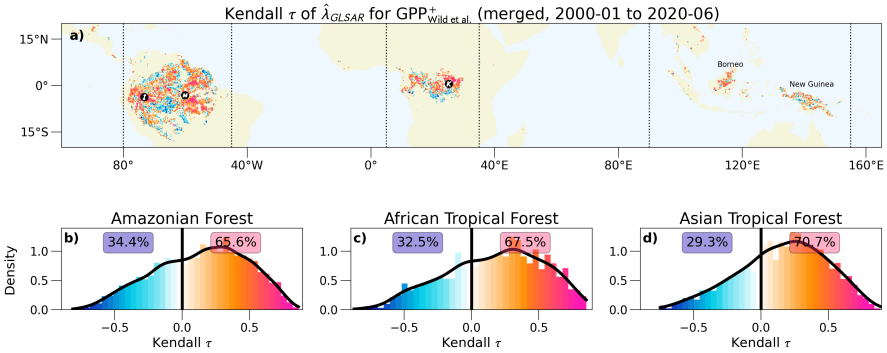

Fig. S2.21 Resilience change accessed as Kendall τ of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for GPP$^{+}_{\mathrm{Wild\ et\ al.}}$. Compare to the linear trend in Fig. 5e-h. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

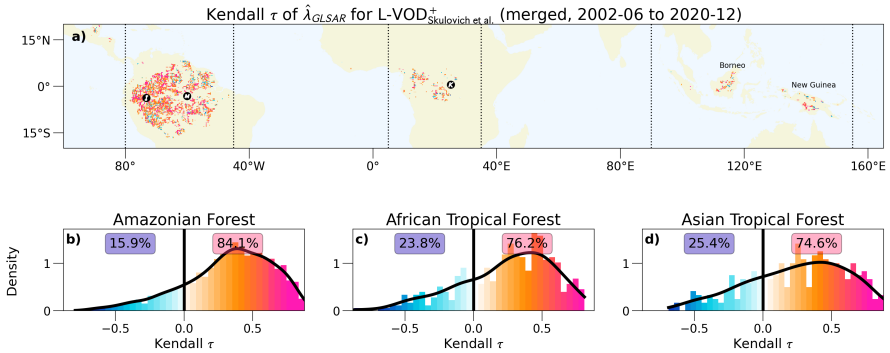

Fig. S2.22 **Resilience change accessed as Kendall τ of $\hat{\lambda}_{\text{GLSAR}}$** calculated on sliding windows of 5 years for L-VOD$^{+}_{\text{Skulovich et al.}}$. Compare to the linear trend in Fig. 5i-l. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

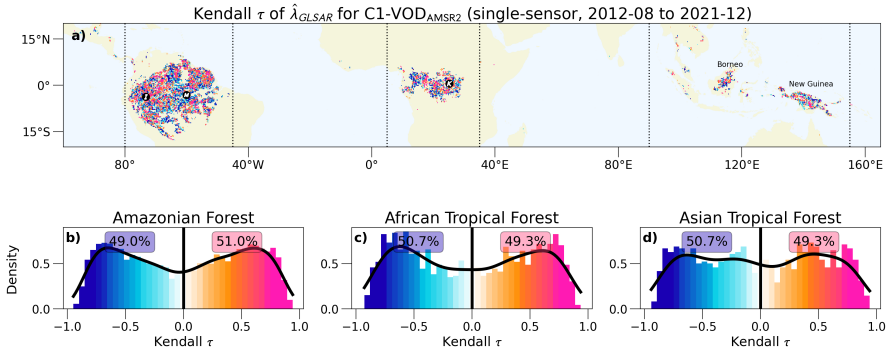

Fig. S2.23 **Resilience change accessed as Kendall τ of $\hat{\lambda}_{\text{GLSAR}}$** calculated on sliding windows of 5 years for C1-VOD$_{\text{AMSR2}}$. Compare to the linear trend in Fig. 4a-d. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

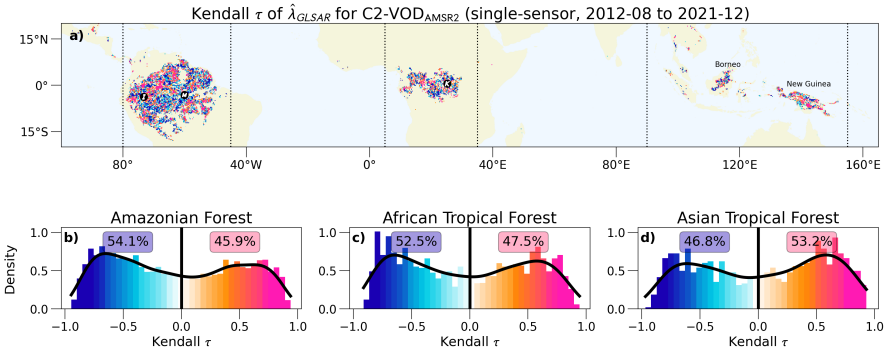

Fig. S2.24 **Resilience change accessed as Kendall τ of $\hat{\lambda}_{\mathrm{GLSAR}}$** calculated on sliding windows of 5 years for C2-VOD$_{\mathrm{AMSR2}}$. Compare to the linear trend in Fig. 4e-h. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

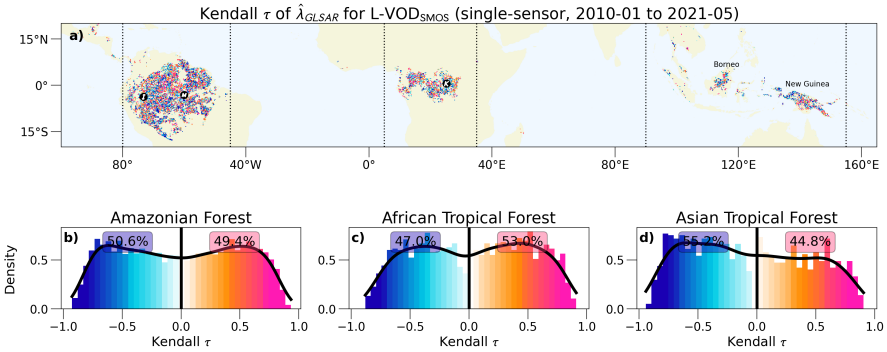

Fig. S2.25 **Resilience change accessed as Kendall τ of $\hat{\lambda}_{\mathrm{GLSAR}}$** calculated on sliding windows of 5 years for L-VOD$_{\mathrm{SMOS}}$. Compare to the linear trend in Fig. 4i-l. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

S2.5 Results for the excluded VPs

In Figs. S2.26 - S2.35 we also depict the linear trends of $\hat{\lambda}_{\mathrm{GLSAR}}$ for all VPs that we excluded from our resilience analysis due to previously gained insights.

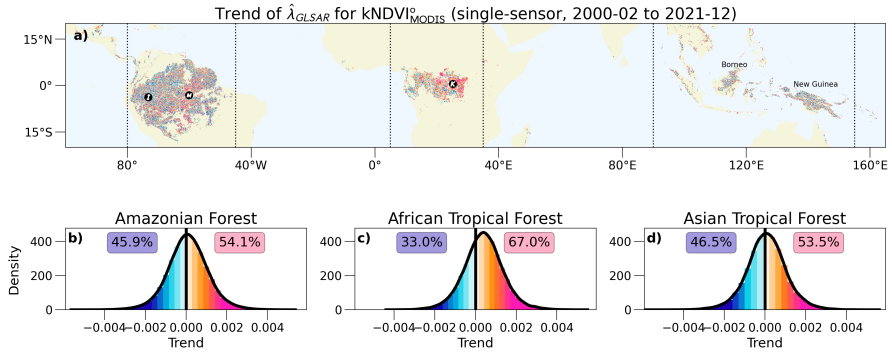

Fig. S2.26 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for kNDVI$^{o}_{\mathrm{MODIS}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

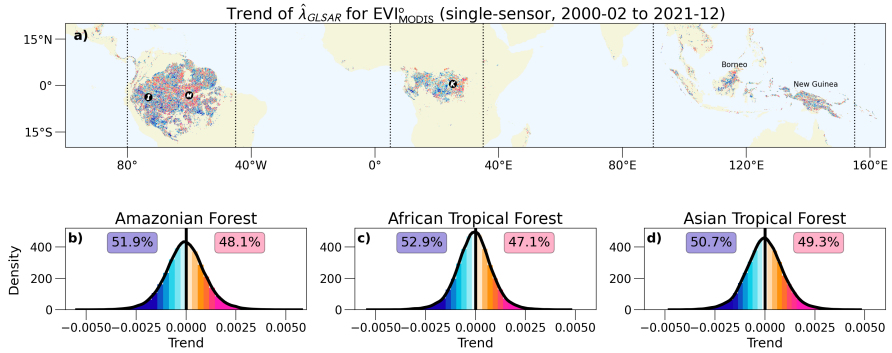

Fig. S2.27 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for EVI$^{o}_{\mathrm{MODIS}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

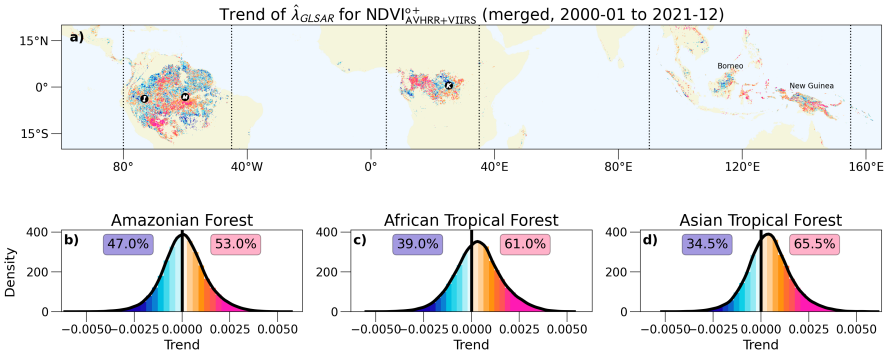

Fig. S2.28 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for $\mathrm{NDVI}^{o+}_{\mathrm{AVHRR+VIIRS}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

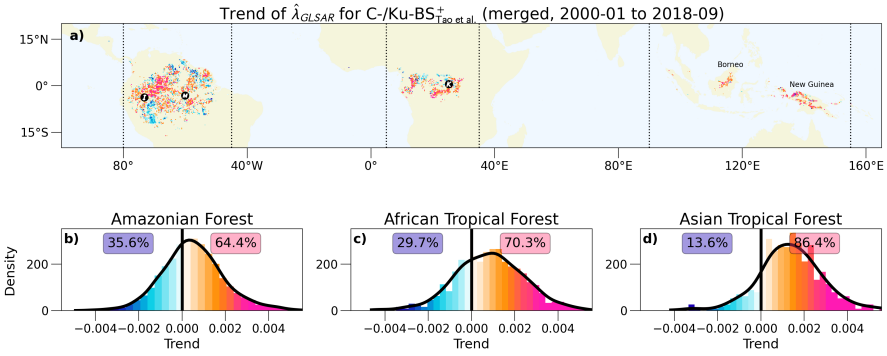

Fig. S2.29 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for $\mathrm{C\text{-}/Ku\text{-}BS}^{+}_{\mathrm{Tao\ et\ al.}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

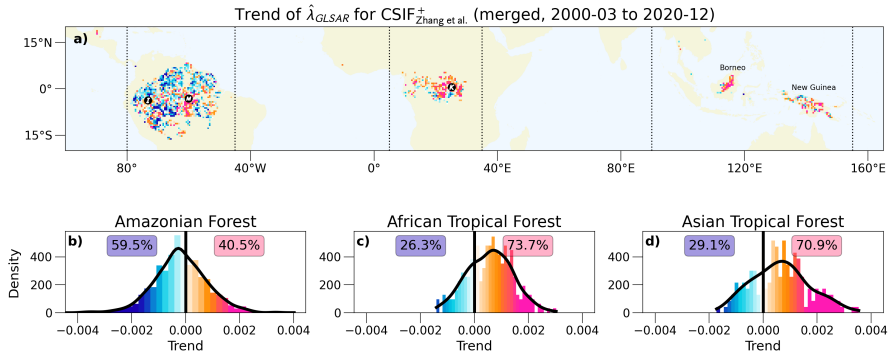

Fig. S2.30 Resilience change accessed as linear trend of $\hat{\lambda}_{\text{GLSAR}}$ calculated on sliding windows of 5 years for SIF$^{+}_{\text{Zhang et al.}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

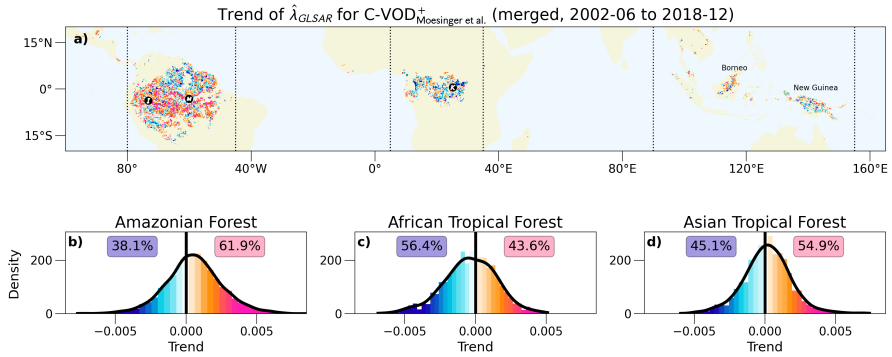

Fig. S2.31 Resilience change accessed as linear trend of $\hat{\lambda}_{\text{GLSAR}}$ calculated on sliding windows of 5 years for C-VOD$^{+}_{\text{Moesinger et al.}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

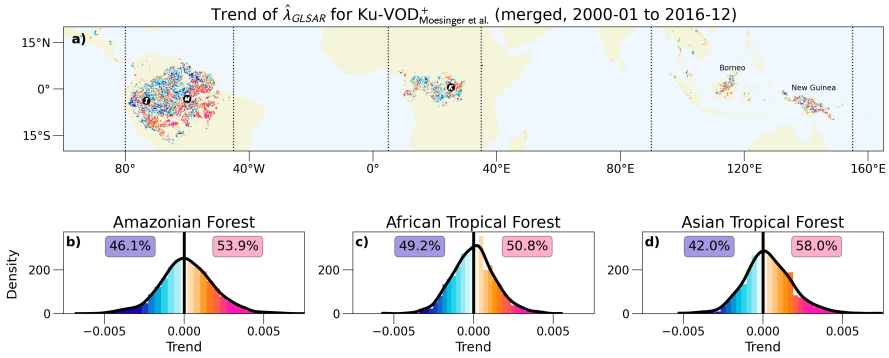

Fig. S2.32 Resilience change accessed as linear trend of $\hat{\lambda}_{\text{GLSAR}}$ calculated on sliding windows of 5 years for Ku-VOD$^{+}_{\text{Moesinger et al.}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

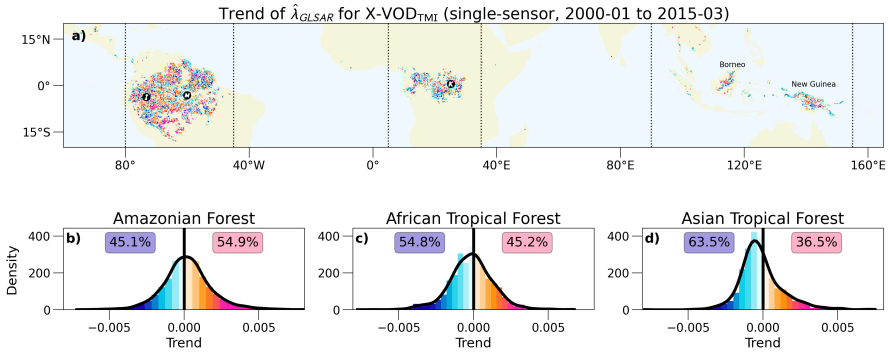

Fig. S2.33 Resilience change accessed as linear trend of $\hat{\lambda}_{\text{GLSAR}}$ calculated on sliding windows of 5 years for X-VOD$_{\text{TMI}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

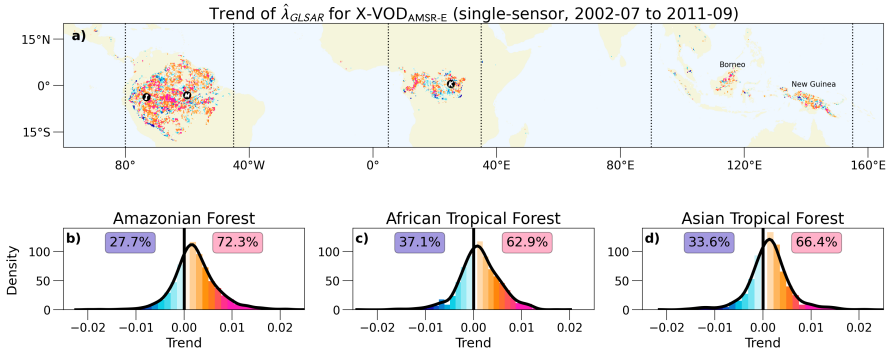

Fig. S2.34 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for X-VOD$_{\mathrm{AMSR\text{-}E}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms.Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

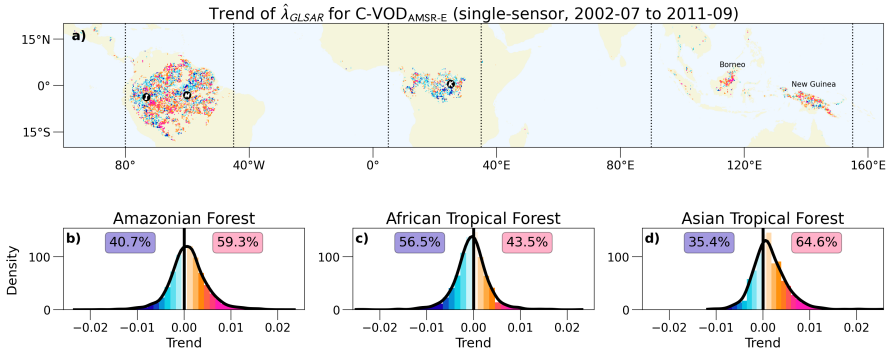

Fig. S2.35 Resilience change accessed as linear trend of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for C-VOD$_{\mathrm{AMSR\text{-}E}}$. The map a) shows the spatial distribution of the linear trend with color-coding equivalent to the corresponding histograms. Compare to Figs. 5 and 4. The cities of Iquitos, Manaus, and Kisangani (black circles from west to east) and the islands Borneo and New Guinea are indicated for reference.

S2.6 Average $\hat{\lambda}_{\mathrm{GLSAR}}$ time series

Fig. S2.36 shows the region-wise $\hat{\lambda}_{\mathrm{GLSAR}}$ time series that are also shown in Fig. 6, however, without them being adjusted to match the range of $\hat{\lambda}_{\mathrm{GLSAR}}$ of $\mathrm{NDVI}^{\circ}_{\mathrm{MODIS}}$.

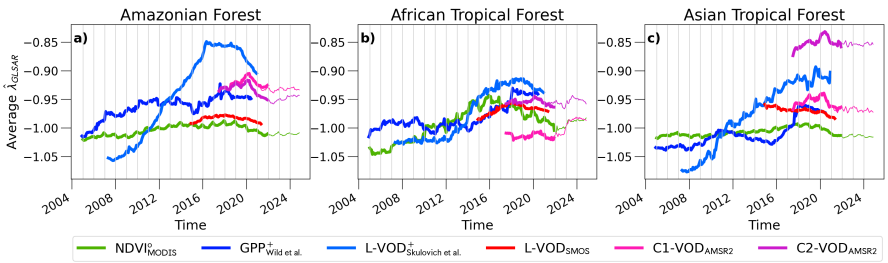

Fig. S2.36 Average time series of $\hat{\lambda}_{\mathrm{GLSAR}}$ calculated on sliding windows of 5 years for a) the Amazon, b) the African and c) the Asian Tropical Forest. For each region, the average $\hat{\lambda}_{\mathrm{GLSAR}}$ is shown for $\mathrm{NDVI}^{\circ}_{\mathrm{MODIS}}$, $\mathrm{GPP}^{+}_{\mathrm{Wild\ et\ al.}}$, $\mathrm{L\text{-}VOD}^{+}_{\mathrm{Skulovich\ et\ al.}}$, $\mathrm{C1\text{-}VOD}_{\mathrm{AMSR2}}$, $\mathrm{C2\text{-}VOD}_{\mathrm{AMSR2}}$, and $\mathrm{L\text{-}VOD}_{\mathrm{SMOS}}$. Compared to Fig. 6, the actual values of $\hat{\lambda}_{\mathrm{GLSAR}}$ are displayed here without adjustment of ranges.

In Fig. S2.37 region-wise $\hat{\lambda}_{\mathrm{GLSAR}}$ time series are shown for all VPs, in particular including those that we excluded from our resilience analysis due to previously gained insights. It is remarkable that the agreement between the VPs we selected due to previously installed criteria is much higher than between the VPs we excluded. This enhances the trust in our method in general, and the results on resilience changes in tropical forests in particular.

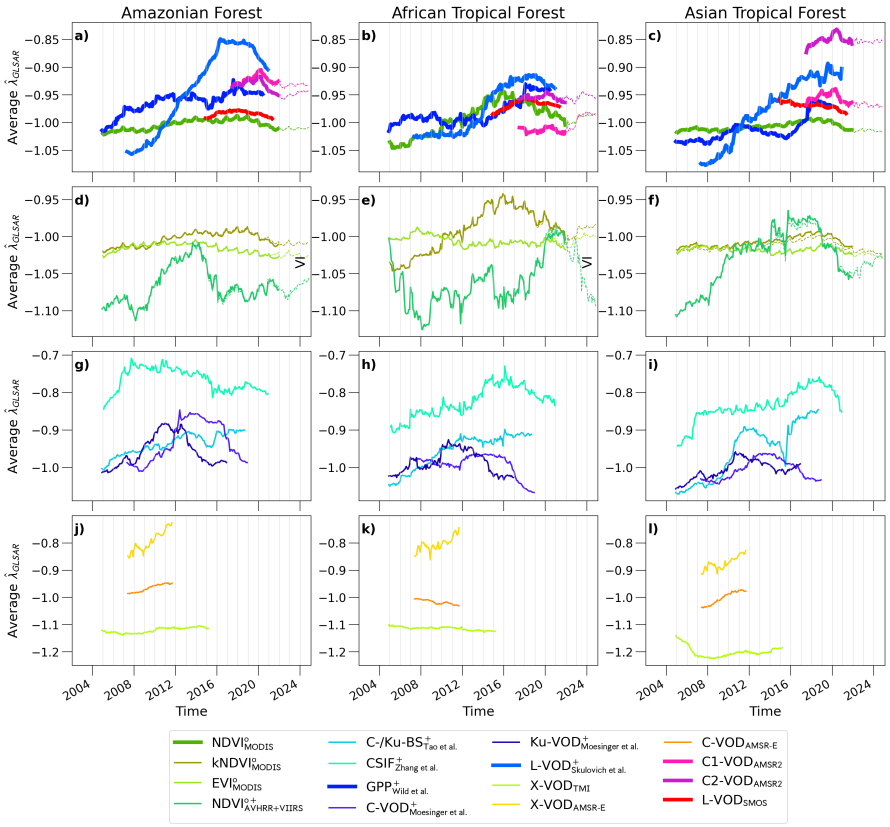

Fig. S2.37 **Average time series of $\hat{\lambda}_{\mathbf{GLSAR}}$** calculated on sliding windows of 5 years for a,d,g,j) the Amazon, b,e,h,k) the African and c,f,i,l) the Asian Tropical Forest. The first row a-c) shows the results on the VPs chosen for the main text. The second row d-f) shows the results for the remaining optical datasets, the third g-i) for the remaining merged datasets ($\text{NDVI}^{o+}_{\text{AVHRR+VIIRS}}$ is shown along with the optical datasets in panels d-f)), and the last j-l) for the excluded single-sensor VOD datasets.

S3 Tables of results

S3.1 Shapiro-Wilk test statistics on the noise time series

VP	median	std	n	
$NDVI^{o}_{MODIS}$	0.9744	0.0326	240105	statistic
	0.0001	0.1508	240105	pvalue
$kNDVI^{o}_{MODIS}$	0.9727	0.0339	240105	statistic
	0.0001	0.1423	240105	pvalue
EVI^{o}_{MODIS}	0.9897	0.0125	240104	statistic
	0.0614	0.2830	240104	pvalue
$NDVI^{o+}_{AVHRR+VIIRS}$	0.9924	0.0092	240031	statistic
	0.1949	0.2999	240031	pvalue
C-/Ku-$BS^{+}_{Tao\ et\ al.}$	0.9899	0.0114	4891	statistic
	0.1181	0.2966	4891	pvalue
$CSIF^{+}_{Zhang\ et\ al.}$	0.9855	0.0155	1645	statistic
	0.0125	0.2148	1645	pvalue
$GPP^{+}_{Wild\ et\ al.}$	0.9876	0.0135	7947	statistic
	0.0332	0.2414	7947	pvalue
C-$VOD^{+}_{Moesinger\ et\ al.}$	0.9890	0.0188	7924	statistic
	0.1334	0.2978	7924	pvalue
Ku-$VOD^{+}_{Moesinger\ et\ al.}$	0.9913	0.0114	7947	statistic
	0.2667	0.3074	7947	pvalue
L-$VOD^{+}_{Skulovich\ et\ al.}$	0.9632	0.0762	4823	statistic
	0.0000	0.1595	4823	pvalue
$X\text{-}VOD_{TMI}$	0.9921	0.0060	8001	statistic
	0.4292	0.3021	8001	pvalue
$X\text{-}VOD_{AMSR\text{-}E}$	0.9888	0.0087	8042	statistic
	0.4988	0.2948	8042	pvalue
$C\text{-}VOD_{AMSR\text{-}E}$	0.9871	0.0126	8042	statistic
	0.3771	0.3109	8042	pvalue
$C1\text{-}VOD_{AMSR2}$	0.9870	0.0186	8001	statistic
	0.3588	0.3139	8001	pvalue
$C2\text{-}VOD_{AMSR2}$	0.9872	0.0150	8001	statistic
	0.3696	0.3093	8001	pvalue
$L\text{-}VOD_{SMOS}$	0.9887	0.0161	9470	statistic
	0.3354	0.3181	9470	pvalue

Table 1: **Results of the Shapiro-Wilk test** on the noise time series in terms of median, standard deviation (std) and number of grid cells used (n). The rows per VP show the Shapiro-Wilk statistic and the p-value, respectively.

S3.2 Breusch-Godfrey test statistics on the noise time series

VP	median	std	n	
$NDVI^{o}_{MODIS}$	33.0402	11.4509	240105	statistic
	0.0003	0.0340	240105	pvalue
$kNDVI^{o}_{MODIS}$	33.0537	11.4514	240105	statistic
	0.0003	0.0338	240105	pvalue
EVI^{o}_{MODIS}	34.4360	11.6626	240104	statistic
	0.0002	0.0315	240104	pvalue
$NDVI^{o+}_{AVHRR+VIIRS}$	30.8539	10.5613	240031	statistic
	0.0006	0.0400	240031	pvalue
$C\text{-/Ku-BS}^{+}_{\text{Tao et al.}}$	29.7779	12.1173	4891	statistic
	0.0009	0.0739	4891	pvalue
$CSIF^{+}_{\text{Zhang et al.}}$	33.8719	13.7625	1645	statistic
	0.0002	0.0718	1645	pvalue
$GPP^{+}_{\text{Wild et al.}}$	56.9296	17.4361	7947	statistic
	0.0000	0.0082	7947	pvalue
$C\text{-VOD}^{+}_{\text{Moesinger et al.}}$	25.8121	10.7026	7924	statistic
	0.0040	0.1092	7924	pvalue
$Ku\text{-VOD}^{+}_{\text{Moesinger et al.}}$	20.9138	9.5460	7947	statistic
	0.0217	0.1545	7947	pvalue
$L\text{-VOD}^{+}_{\text{Skulovich et al.}}$	26.7159	12.3816	4823	statistic
	0.0029	0.1038	4823	pvalue
$X\text{-VOD}_{TMI}$	33.9847	25.2926	8001	statistic
	0.0002	0.0517	8001	pvalue
$X\text{-VOD}_{AMSR\text{-}E}$	17.3514	7.6879	8042	statistic
	0.0669	0.2065	8042	pvalue
$C\text{-VOD}_{AMSR\text{-}E}$	19.6450	8.3805	8042	statistic
	0.0328	0.1729	8042	pvalue
$C1\text{-VOD}_{AMSR2}$	19.0406	8.1729	8001	statistic
	0.0397	0.1784	8001	pvalue
$C2\text{-VOD}_{AMSR2}$	18.9028	8.0777	8001	statistic
	0.0415	0.1878	8001	pvalue
$L\text{-VOD}_{SMOS}$	40.8155	16.7825	9470	statistic
	0.0000	0.0593	9470	pvalue

Table 2: **Results of the Breusch-Godfrey test** on the noise time series in terms of median, standard deviation (std) and number of grid cells used (n). The rows per VP show the Breusch-Godfrey statistic and the p-value, respectively.

S3.3 Kendall τ statistics of σ of the noise over time

VP	median	std	n
$NDVI^{o}_{MODIS}$	-0.0840	0.3305	3160
$kNDVI^{o}_{MODIS}$	-0.0908	0.3294	3187

EVI^{o}_{MODIS}	0.0040	0.3305	1782
$NDVI^{o+}_{AVHRR+VIIRS}$	0.2233	0.2608	365
C-/Ku-$BS^{+}_{Tao\ et\ al.}$	-0.5219	0.3608	510
$CSIF^{+}_{Zhang\ et\ al.}$	-0.1044	0.3420	1045
$GPP^{+}_{Wild\ et\ al.}$	0.2222	0.3027	591
C-$VOD^{+}_{Moesinger\ et\ al.}$	0.2163	0.3977	1002
Ku-$VOD^{+}_{Moesinger\ et\ al.}$	0.2141	0.3522	612
L-$VOD^{+}_{Skulovich\ et\ al.}$	0.0648	0.4717	225
X-VOD_{TMI}	-0.1812	0.3829	165
X-$VOD_{AMSR\text{-}E}$	-0.0633	0.4984	5878
C-$VOD_{AMSR\text{-}E}$	-0.0596	0.5093	2196
C1-VOD_{AMSR2}	0.1796	0.5219	2865
C2-VOD_{AMSR2}	0.0678	0.5253	3027
L-VOD_{SMOS}	-0.0030	0.5302	2003

Table 3: **Kendall τ of the noise amplitude** calculated on sliding windows. The columns state the median, standard deviation (std) and number of grid cells used (n).

S3.4 Correlation statistics of $\lambda_{empirical}$ with theoretical metrics

VP	R^2	$\hat{\lambda}_{GLSAR}$	λ_{GLSAR}	λ_{AC1}	$\lambda_{Var/\sigma(LR)^2}$	AC1	
$NDVI^{o}_{MODIS}$	0.1	0.0764	0.0632	0.0530	0.0290	0.0775	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		111038	64657	64200	76138	111038	n
	0.2	0.1234	0.0910	0.0788	0.0496	0.1237	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		67574	41486	41201	46923	67574	n
	0.3	0.1716	0.1235	0.1076	0.0779	0.1729	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		35759	22945	22807	25047	35759	n
	0.4	0.2343	0.1692	0.1577	0.1245	0.2373	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		16782	11116	11047	11877	16782	n
	0.5	0.2996	0.2277	0.2072	0.1783	0.3041	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		6731	4534	4509	4829	6731	n
$kNDVI^{o}_{MODIS}$	0.1	0.0784	0.0624	0.0543	0.0293	0.0793	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		112018	65285	64790	77056	112018	n
	0.2	0.1250	0.0893	0.0784	0.0484	0.1251	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p

		69528	42693	42382	48394	69528	n
	0.3	0.1728	0.1253	0.1093	0.0752	0.1742	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		37574	24092	23941	26432	37574	n
	0.4	0.2376	0.1708	0.1574	0.1222	0.2400	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		18068	11988	11908	12833	18068	n
	0.5	0.2963	0.2072	0.1953	0.1542	0.3020	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		7349	4982	4947	5271	7349	n
EVI^{o}_{MODIS}	0.1	0.0552	0.0474	0.0405	0.0064	0.0587	c
		0.0000	0.0000	0.0000	0.1765	0.0000	p
		67465	35090	34937	44763	67465	n
	0.2	0.0863	0.0698	0.0656	0.0284	0.0894	c
		0.0000	0.0000	0.0000	0.0006	0.0000	p
		21370	12103	12030	14434	21370	n
	0.3	0.1292	0.0831	0.0810	0.0377	0.1324	c
		0.0000	0.0000	0.0000	0.0150	0.0000	p
		6027	3599	3578	4157	6027	n
	0.4	0.2095	0.1364	0.1286	0.0741	0.2062	c
		0.0000	0.0000	0.0001	0.0131	0.0000	p
		1591	958	963	1119	1591	n
	0.5	0.3279	0.2482	0.2283	0.0229	0.3360	c
		0.0000	0.0003	0.0009	0.7273	0.0000	p
		321	207	207	235	321	n
$NDVI^{o+}_{AVHRR+VIIRS}$	0.1	0.0690	0.0557	0.0337	-0.0538	0.0907	c
		0.0000	0.0000	0.0022	0.0000	0.0000	p
		29422	8297	8248	18967	29422	n
	0.2	0.0737	0.0783	0.0632	-0.0216	0.1162	c
		0.0000	0.0064	0.0278	0.3159	0.0000	p
		3522	1212	1211	2164	3522	n
	0.3	0.0665	0.1046	0.0985	0.1324	0.1158	c
		0.2145	0.1748	0.2001	0.0560	0.0303	p
		350	170	171	209	350	n
	0.4	0.0123	0.5801	0.5946	0.0134	0.0922	c
		0.9545	0.0146	0.0118	0.9636	0.6684	p
		24	17	17	14	24	n
	0.5	nan	nan	nan	nan	nan	c
		nan	nan	nan	nan	nan	p
		0	0	0	0	0	n
$C\text{-}/Ku\text{-}BS^{+}_{\text{Tao et al.}}$	0.1	0.0388	0.0739	0.0390	0.0389	0.0229	c

		0.0724	0.0033	0.1217	0.1095	0.2892	p
		2141	1581	1575	1690	2141	n
	0.2	0.0251	0.0275	-0.0225	-0.0004	-0.0092	c
		0.4210	0.4284	0.5178	0.9900	0.7670	p
		1029	830	825	861	1029	n
	0.3	0.0578	0.0572	-0.0030	0.0564	0.0068	c
		0.2812	0.3142	0.9574	0.3316	0.8985	p
		350	312	310	298	350	n
	0.4	-0.0237	-0.0673	-0.1356	0.0282	-0.0764	c
		0.8345	0.5660	0.2493	0.8131	0.5008	p
		80	75	74	73	80	n
	0.5	-0.2893	0.3467	0.7712	0.7619	-0.1682	c
		0.4870	0.4461	0.0423	0.0465	0.6905	p
		8	7	7	7	8	n
$CSIF^{+}_{\text{Zhang et al.}}$	0.1	0.0545	0.1304	0.1020	0.0682	0.0719	c
		0.1603	0.0008	0.0087	0.0800	0.0640	p
		665	662	661	661	665	n
	0.2	0.0785	0.1234	0.1436	0.1116	0.1052	c
		0.1660	0.0293	0.0111	0.0496	0.0630	p
		313	312	312	310	313	n
	0.3	0.0936	0.0769	0.0979	0.1853	0.1356	c
		0.3131	0.4077	0.2918	0.0455	0.1431	p
		118	118	118	117	118	n
	0.4	0.2112	0.2055	0.2842	0.3007	0.3140	c
		0.2626	0.2761	0.1280	0.1064	0.0911	p
		30	30	30	30	30	n
	0.5	0.2814	0.1624	0.1644	0.1668	0.2273	c
		0.4632	0.6763	0.6725	0.6680	0.5564	p
		9	9	9	9	9	n
$GPP^{+}_{\text{Wild et al.}}$	0.1	0.1959	0.1561	0.1637	0.1199	0.1975	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		2477	1917	1912	1759	2477	n
	0.2	0.2154	0.1370	0.1312	0.1842	0.2063	c
		0.0000	0.0001	0.0001	0.0000	0.0000	p
		1054	863	861	783	1054	n
	0.3	0.3293	0.2381	0.2145	0.2646	0.3100	c
		0.0000	0.0000	0.0001	0.0000	0.0000	p
		409	342	342	313	409	n
	0.4	0.5801	0.5310	0.5203	0.5805	0.5769	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		147	132	132	112	147	n

	0.5	0.7538	0.6583	0.6776	0.7829	0.7537	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		46	42	42	36	46	n
C-VOD$^{+}_{\text{Moesinger et al.}}$	0.1	0.2415	0.1979	0.2112	0.2124	0.2539	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		2095	1487	1481	1704	2095	n
	0.2	0.2868	0.2284	0.2222	0.2293	0.3039	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		1121	863	861	957	1121	n
	0.3	0.3552	0.2611	0.2731	0.2866	0.3654	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		487	400	401	434	487	n
	0.4	0.3369	0.1580	0.1849	0.3271	0.4012	c
		0.0000	0.0568	0.0255	0.0000	0.0000	p
		166	146	146	151	166	n
	0.5	0.3244	-0.0536	0.0190	0.3883	0.4385	c
		0.0860	0.7948	0.9264	0.0411	0.0173	p
		29	26	26	28	29	n
Ku-VOD$^{+}_{\text{Moesinger et al.}}$	0.1	0.2327	0.1400	0.0459	0.1503	0.2192	c
		0.0000	0.0030	0.3323	0.0012	0.0000	p
		616	447	449	462	616	n
	0.2	0.2926	0.1480	-0.0200	0.0960	0.2437	c
		0.0001	0.0727	0.8086	0.2557	0.0014	p
		169	148	149	142	169	n
	0.3	0.2469	0.0603	-0.0988	0.0813	0.1104	c
		0.0719	0.6709	0.4858	0.5745	0.4268	p
		54	52	52	50	54	n
	0.4	0.1470	0.0474	-0.0815	-0.0521	0.0014	c
		0.5364	0.8426	0.7327	0.8321	0.9955	p
		20	20	20	19	20	n
	0.5	0.4917	0.5019	0.2871	0.2873	0.2984	c
		0.2159	0.2050	0.4906	0.4902	0.4728	p
		8	8	8	8	8	n
L-VOD$^{+}_{\text{Skulovich et al.}}$	0.1	0.1642	0.1588	0.1075	0.1335	0.1435	c
		0.0000	0.0000	0.0001	0.0000	0.0000	p
		1710	1386	1382	1339	1710	n
	0.2	0.2174	0.1736	0.1114	0.1640	0.1850	c
		0.0000	0.0000	0.0013	0.0000	0.0000	p
		1003	827	827	807	1003	n
	0.3	0.2615	0.2014	0.1373	0.1830	0.2169	c
		0.0000	0.0000	0.0042	0.0002	0.0000	p

		508	433	434	418	508	n
	0.4	0.3893	0.2888	0.2212	0.3489	0.3460	c
		0.0000	0.0000	0.0012	0.0000	0.0000	p
		238	212	213	206	238	n
	0.5	0.4880	0.3880	0.3142	0.3826	0.4525	c
		0.0000	0.0001	0.0012	0.0001	0.0000	p
		114	103	104	102	114	n
X-VOD$_{\text{TMI}}$	0.1	0.2177	0.1506	0.1489	-0.0502	0.2272	c
		0.0000	0.0226	0.0242	0.1261	0.0000	p
		1118	229	229	930	1118	n
	0.2	0.3896	0.2917	0.2494	-0.0200	0.3999	c
		0.0000	0.0032	0.0123	0.7659	0.0000	p
		283	100	100	225	283	n
	0.3	0.5196	0.4962	0.4344	0.2570	0.5334	c
		0.0000	0.0008	0.0040	0.0582	0.0000	p
		73	42	42	55	73	n
	0.4	0.0018	-0.2220	-0.2295	-0.1360	-0.0061	c
		0.9950	0.5117	0.4973	0.7079	0.9829	p
		15	11	11	10	15	n
	0.5	-0.9993	-0.9976	-0.9936	nan	-0.9986	c
		0.0231	0.0440	0.0722	nan	0.0341	p
		3	3	3	2	3	n
X-VOD$_{\text{AMSR-E}}$	0.1	0.1457	0.1135	0.1778	0.1815	0.1853	c
		0.0020	0.0195	0.0002	0.0002	0.0001	p
		448	424	423	422	448	n
	0.2	0.2270	0.2460	0.3320	0.2926	0.3252	c
		0.0057	0.0034	0.0001	0.0005	0.0001	p
		147	140	140	138	147	n
	0.3	0.3153	0.3074	0.1449	-0.0366	0.0973	c
		0.0650	0.0725	0.4062	0.8371	0.5781	p
		35	35	35	34	35	n
	0.4	0.7300	0.7865	0.9090	0.8242	0.8117	c
		0.0995	0.0635	0.0120	0.0861	0.0498	p
		6	6	6	5	6	n
	0.5	nan	nan	nan	nan	nan	c
		nan	nan	nan	nan	nan	p
		1	1	1	1	1	n
C-VOD$_{\text{AMSR-E}}$	0.1	0.1070	0.0862	0.0422	0.0344	0.0954	c
		0.0034	0.0699	0.3756	0.4442	0.0091	p
		747	443	443	497	747	n
	0.2	0.1383	0.1728	0.0682	0.0918	0.1257	c

		0.0334	0.0279	0.3882	0.2325	0.0533	p
		237	162	162	171	237	n
	0.3	0.2219	0.2056	0.0230	0.2648	0.1640	c
		0.0972	0.1861	0.8822	0.0720	0.2227	p
		57	43	44	47	57	n
	0.4	0.3023	0.8019	0.7392	0.4278	0.2641	c
		0.2734	0.0030	0.0093	0.1448	0.3414	p
		15	11	11	13	15	n
	0.5	-0.6619	-0.7222	-0.8633	-0.8393	-0.8401	c
		0.5395	0.4862	0.3368	0.3660	0.3650	p
		3	3	3	3	3	n
C1-VOD$_{\text{AMSR2}}$	0.1	0.2704	0.2298	0.1987	0.2516	0.2990	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		1179	813	813	833	1179	n
	0.2	0.4278	0.3603	0.3591	0.3428	0.4488	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		400	310	308	309	400	n
	0.3	0.5162	0.3810	0.3621	0.4505	0.5579	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		140	124	124	131	140	n
	0.4	0.5849	0.4036	0.4796	0.5760	0.6729	c
		0.0000	0.0120	0.0023	0.0001	0.0000	p
		43	38	38	41	43	n
	0.5	0.5551	0.6980	0.9072	0.9064	0.8573	c
		0.0958	0.0248	0.0003	0.0003	0.0015	p
		10	10	10	10	10	n
C2-VOD$_{\text{AMSR2}}$	0.1	0.2382	0.2345	0.1969	0.2002	0.2320	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		1045	730	731	707	1045	n
	0.2	0.3821	0.3640	0.3407	0.3121	0.3779	c
		0.0000	0.0000	0.0000	0.0000	0.0000	p
		385	303	303	298	385	n
	0.3	0.5258	0.4380	0.4349	0.3326	0.5281	c
		0.0000	0.0000	0.0000	0.0003	0.0000	p
		129	113	113	112	129	n
	0.4	0.7231	0.6480	0.6504	0.6156	0.7315	c
		0.0000	0.0000	0.0000	0.0001	0.0000	p
		36	33	33	34	36	n
	0.5	0.7815	0.7399	0.6036	0.6095	0.6027	c
		0.0664	0.0927	0.2045	0.1990	0.2054	p
		6	6	6	6	6	n

L-VOD$_{\text{SMOS}}$	0.1	0.0691	0.0916	0.0970	0.0529	0.0791	c
		0.0155	0.0113	0.0076	0.1223	0.0056	p
		1228	763	756	854	1228	n
	0.2	0.2065	0.1888	0.1751	0.1041	0.2314	c
		0.0001	0.0027	0.0057	0.0849	0.0000	p
		369	250	248	275	369	n
	0.3	0.2343	0.0477	0.0521	-0.0055	0.2515	c
		0.0133	0.6764	0.6464	0.9626	0.0077	p
		111	79	80	76	111	n
	0.4	0.5346	0.2117	0.2331	0.3916	0.5370	c
		0.0028	0.3569	0.3092	0.0646	0.0027	p
		29	21	21	23	29	n
	0.5	0.7463	0.3020	0.2754	0.5694	0.7135	c
		0.0083	0.5103	0.5500	0.1822	0.0137	p
		11	7	7	7	11	n

Table 4: **Pearson correlation results of $\lambda_{\text{empirical}}$ with the five best theoretical metrics**, i.e. $\hat{\lambda}_{\text{GLSAR}}$, λ_{GLSAR}, λ_{AC1}, λ_{LR}, and AC1 (columns) for all VPs (rows) and R^2 values of the $\lambda_{\text{empirical}}$ (sub-rows). The right-most c, p, n indicate correlation coefficient, p-value, and number of grid cells taken into account for the correlation, respectively.